Supplementary material: A public-health and clinical tabular benchmark for target-domain recalibration of conformal prediction under deployment shift

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Sections and tables have separate numbering. Sections S1–S13 describe the additional outcome-blind protocol on the three public datasets; S14–S16 describe independent institutional cohorts. Section S17 contrasts the primary and additional protocols and supplies primary label counts. Section S18 reports the detailed primary-protocol evidence. Protocol-specific results are not pooled. Section S19 supplies detailed comparator implementations and data preparation; S20 collects additional protocol, predictor, label-observation and institutional summaries. The later material is located as follows:

Section	Material	Tables / figure
S9	Predictor family	Table S10
S10	Feature ablation and L2	Tables S11–S12
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S1. Absolute and class-specific label budgets

There are 25 seed–draw combinations for each condition and fraction. Positive class rank-overflow counts include absent classes and indicate an infinite threshold. All fractions use the full target-domain denominator.

Table S1: Absolute target labels and per-class ranges across five seeds and five draws.

Data	Condition	Fraction	Labels	Class 0	Class 1	Overflow 1
BRFSS	Age 65+	0.5%	727	549–592	135–178	0
BRFSS	Age 65+	1%	1453	1111–1161	292–342	0
BRFSS	Age 65+	2%	2907	2227–2309	598–680	0
BRFSS	Age 65+	3%	4360	3374–3466	894–986	0
BRFSS	Age 65+	5%	7267	5649–5792	1475–1618	0
BRFSS	Age 65+	10%	14534	11375–11501	3033–3159	0
BRFSS	Age 65+	15%	21801	17090–17222	4579–4711	0
BRFSS	Age 65+	20%	29069	22821–22944	6125–6248	0
BRFSS	BMI threshold	0.5%	589	430–462	127–159	0
BRFSS	BMI threshold	1%	1178	859–931	247–319	0
BRFSS	BMI threshold	2%	2356	1752–1837	519–604	0
BRFSS	BMI threshold	3%	3534	2655–2752	782–879	0
BRFSS	BMI threshold	5%	5891	4437–4561	1330–1454	0
BRFSS	BMI threshold	10%	11781	8883–9074	2707–2898	0
BRFSS	BMI threshold	15%	17672	13364–13537	4135–4308	0
BRFSS	BMI threshold	20%	23562	17866–17996	5566–5696	0
BRFSS	Race proxy	0.5%	137	99–116	21–38	0
BRFSS	Race proxy	1%	274	207–223	51–67	0
BRFSS	Race proxy	2%	548	422–447	101–126	0
BRFSS	Race proxy	3%	823	627–668	155–196	0
BRFSS	Race proxy	5%	1371	1039–1114	257–332	0
BRFSS	Race proxy	10%	2742	2119–2201	541–623	0
BRFSS	Race proxy	15%	4113	3180–3287	826–933	0
BRFSS	Race proxy	20%	5484	4265–4365	1119–1219	0
BRFSS	South region	0.5%	610	495–525	85–115	0
BRFSS	South region	1%	1220	1007–1045	175–213	0
BRFSS	South region	2%	2440	2019–2074	366–421	0
BRFSS	South region	3%	3660	3042–3124	536–618	0
BRFSS	South region	5%	6100	5061–5198	902–1039	0
BRFSS	South region	10%	12200	10172–10326	1874–2028	0
BRFSS	South region	15%	18300	15240–15419	2881–3060	0
BRFSS	South region	20%	24400	20362–20538	3862–4038	0
BRFSS	Later years	0.5%	1200	1000–1049	151–200	0
BRFSS	Later years	1%	2400	2031–2090	310–369	0
BRFSS	Later years	2%	4800	4094–4161	639–706	0
BRFSS	Later years	3%	7200	6124–6230	970–1076	0
BRFSS	Later years	5%	12000	10259–10390	1610–1741	0
BRFSS	Later years	10%	24000	20554–20731	3269–3446	0
BRFSS	Later years	15%	36000	30846–31047	4953–5154	0
BRFSS	Later years	20%	48000	41136–41386	6614–6864	0
NHANES	Age 65+	0.5%	26	11–21	5–15	16

Data	Condition	Fraction	Labels	Class 0	Class 1	Overflow 1
NHANES	Age 65+	1%	52	31–43	9–21	0
NHANES	Age 65+	2%	104	67–81	23–37	0
NHANES	Age 65+	3%	156	98–119	37–58	0
NHANES	Age 65+	5%	260	172–199	61–88	0
NHANES	Age 65+	10%	521	362–384	137–159	0
NHANES	Age 65+	15%	781	546–574	207–235	0
NHANES	Age 65+	20%	1042	729–769	273–313	0
NHANES	BMI threshold	0.5%	45	31–42	3–14	19
NHANES	BMI threshold	1%	89	68–79	10–21	0
NHANES	BMI threshold	2%	178	139–152	26–39	0
NHANES	BMI threshold	3%	268	212–230	38–56	0
NHANES	BMI threshold	5%	446	354–374	72–92	0
NHANES	BMI threshold	10%	892	711–745	147–181	0
NHANES	BMI threshold	15%	1338	1073–1108	230–265	0
NHANES	BMI threshold	20%	1784	1433–1467	317–351	0
NHANES	Race proxy	0.5%	43	36–41	2–7	25
NHANES	Race proxy	1%	87	73–83	4–14	10
NHANES	Race proxy	2%	174	151–164	10–23	0
NHANES	Race proxy	3%	261	227–249	12–34	0
NHANES	Race proxy	5%	435	383–403	32–52	0
NHANES	Race proxy	10%	870	772–797	73–98	0
NHANES	Race proxy	15%	1305	1160–1193	112–145	0
NHANES	Race proxy	20%	1740	1550–1585	155–190	0
NHANES	Later years	0.5%	91	76–88	3–15	11
NHANES	Later years	1%	181	151–172	9–30	0
NHANES	Later years	2%	363	312–340	23–51	0
NHANES	Later years	3%	544	471–507	37–73	0
NHANES	Later years	5%	907	801–849	58–106	0
NHANES	Later years	10%	1813	1621–1658	155–192	0
NHANES	Later years	15%	2720	2433–2472	248–287	0
NHANES	Later years	20%	3627	3249–3302	325–378	0
UCI	Age 70+	0.5%	158	134–148	10–24	0
UCI	Age 70+	1%	317	270–295	22–47	0
UCI	Age 70+	2%	634	544–585	49–90	0
UCI	Age 70+	3%	951	822–872	79–129	0
UCI	Age 70+	5%	1585	1394–1445	140–191	0
UCI	Age 70+	10%	3170	2828–2887	283–342	0
UCI	Age 70+	15%	4755	4248–4317	438–507	0
UCI	Age 70+	20%	6339	5675–5735	604–664	0
UCI	Admission source	0.5%	191	161–182	9–30	0
UCI	Admission source	1%	383	336–357	26–47	0
UCI	Admission source	2%	766	681–710	56–85	0
UCI	Admission source	3%	1149	1022–1063	86–127	0

Data	Condition	Fraction	Labels	Class 0	Class 1	Overflow 1
UCI	Admission source	5%	1914	1718–1764	150–196	0
UCI	Admission source	10%	3829	3430–3533	296–399	0
UCI	Admission source	15%	5743	5172–5271	472–571	0
UCI	Admission source	20%	7658	6925–7017	641–733	0
UCI	Race proxy	0.5%	64	53–63	1–11	23
UCI	Race proxy	1%	129	111–126	3–18	5
UCI	Race proxy	2%	258	226–245	13–32	0
UCI	Race proxy	3%	387	344–365	22–43	0
UCI	Race proxy	5%	644	578–597	47–66	0
UCI	Race proxy	10%	1289	1166–1195	94–123	0
UCI	Race proxy	15%	1933	1755–1778	155–178	0
UCI	Race proxy	20%	2577	2348–2373	204–229	0

S2. Demographic subgroup results

Counts refer to untouched target-test observations and are ranges over seeds. Coverage and size average draws and seeds. The NHANES age group below 65 includes children as well as adults. No class-conditional guarantee is asserted for demographic subgroups.

Table S2: Target-test subgroups at the 10% target-label setting. Counts are seed ranges; prevalences and metrics are means.

Data	Condition	Method	Variable	Subgroup	N	Prevalence	Coverage	Size
BRFSS	Age 65+	S. split	Age	65_plus	101741–101741	0.213	0.809	1.448
BRFSS	Age 65+	S. split	Sex	female	60753–60997	0.197	0.810	1.443
BRFSS	Age 65+	S. split	Sex	male	40732–40979	0.236	0.808	1.457
BRFSS	Age 65+	S. split	Sex	unknown	7–12	0.274	0.807	1.442
BRFSS	Age 65+	T. Mondrian	Age	65_plus	101741–101741	0.213	0.901	1.511
BRFSS	Age 65+	T. Mondrian	Sex	female	60753–60997	0.197	0.903	1.501
BRFSS	Age 65+	T. Mondrian	Sex	male	40732–40979	0.236	0.898	1.527
BRFSS	Age 65+	T. Mondrian	Sex	unknown	7–12	0.274	0.830	1.423
BRFSS	Age 65+	T. split	Age	65_plus	101741–101741	0.213	0.901	1.660
BRFSS	Age 65+	T. split	Sex	female	60753–60997	0.197	0.901	1.654
BRFSS	Age 65+	T. split	Sex	male	40732–40979	0.236	0.900	1.670
BRFSS	Age 65+	T. split	Sex	unknown	7–12	0.274	0.853	1.504
BRFSS	BMI threshold	S. split	Age	18_64	55814–56029	0.183	0.745	1.387
BRFSS	BMI threshold	S. split	Age	65_plus	25976–26168	0.355	0.541	1.202
BRFSS	BMI threshold	S. split	Age	unknown	463–498	0.218	0.689	1.436
BRFSS	BMI threshold	S. split	Sex	female	44504–44612	0.239	0.657	1.317
BRFSS	BMI threshold	S. split	Sex	male	37850–37957	0.236	0.708	1.342
BRFSS	BMI threshold	S. split	Sex	unknown	4–7	0.410	0.645	1.410

Data	Condition	Method	Variable	Subgroup	N	Prevalence	Coverage	Size
BRFSS	BMI threshold	T. Mondrian	Age	18_64	55814–56029	0.183	0.920	1.356
BRFSS	BMI threshold	T. Mondrian	Age	65_plus	25976–26168	0.355	0.853	1.568
BRFSS	BMI threshold	T. Mondrian	Age	unknown	463–498	0.218	0.929	1.505
BRFSS	BMI threshold	T. Mondrian	Sex	female	44504–44612	0.239	0.891	1.435
BRFSS	BMI threshold	T. Mondrian	Sex	male	37850–37957	0.236	0.909	1.410
BRFSS	BMI threshold	T. Mondrian	Sex	unknown	4–7	0.410	0.629	1.288
BRFSS	BMI threshold	T. split	Age	18_64	55814–56029	0.183	0.931	1.801
BRFSS	BMI threshold	T. split	Age	65_plus	25976–26168	0.355	0.829	1.642
BRFSS	BMI threshold	T. split	Age	unknown	463–498	0.218	0.932	1.855
BRFSS	BMI threshold	T. split	Sex	female	44504–44612	0.239	0.886	1.738
BRFSS	BMI threshold	T. split	Sex	male	37850–37957	0.236	0.914	1.767
BRFSS	BMI threshold	T. split	Sex	unknown	4–7	0.410	0.766	1.530
BRFSS	Race proxy	S. split	Age	18_64	13376–13483	0.149	0.912	1.363
BRFSS	Race proxy	S. split	Age	65_plus	5402–5504	0.362	0.746	1.441
BRFSS	Race proxy	S. split	Age	unknown	298–321	0.152	0.929	1.554
BRFSS	Race proxy	S. split	Sex	female	11691–11758	0.218	0.855	1.397
BRFSS	Race proxy	S. split	Sex	male	7435–7501	0.196	0.880	1.374
BRFSS	Race proxy	S. split	Sex	unknown	1–3	0.300	1.000	1.300
BRFSS	Race proxy	T. Mondrian	Age	18_64	13376–13483	0.149	0.932	1.298
BRFSS	Race proxy	T. Mondrian	Age	65_plus	5402–5504	0.362	0.817	1.517
BRFSS	Race proxy	T. Mondrian	Age	unknown	298–321	0.152	0.946	1.436
BRFSS	Race proxy	T. Mondrian	Sex	female	11691–11758	0.218	0.894	1.376
BRFSS	Race proxy	T. Mondrian	Sex	male	7435–7501	0.196	0.909	1.342
BRFSS	Race proxy	T. Mondrian	Sex	unknown	1–3	0.300	1.000	1.200
BRFSS	Race proxy	T. split	Age	18_64	13376–13483	0.149	0.936	1.447
BRFSS	Race proxy	T. split	Age	65_plus	5402–5504	0.362	0.804	1.551
BRFSS	Race proxy	T. split	Age	unknown	298–321	0.152	0.958	1.656
BRFSS	Race proxy	T. split	Sex	female	11691–11758	0.218	0.891	1.492
BRFSS	Race proxy	T. split	Sex	male	7435–7501	0.196	0.912	1.462
BRFSS	Race proxy	T. split	Sex	unknown	1–3	0.300	1.000	1.300
BRFSS	South region	S. split	Age	18_64	52985–53195	0.116	0.913	1.317
BRFSS	South region	S. split	Age	65_plus	30978–31185	0.242	0.789	1.477
BRFSS	South region	S. split	Age	unknown	1216–1249	0.127	0.924	1.498
BRFSS	South region	S. split	Sex	female	49770–49982	0.159	0.864	1.380
BRFSS	South region	S. split	Sex	male	35412–35627	0.166	0.873	1.376

Data	Condition	Method	Variable	Subgroup	*N*	Prevalence	Coverage	Size
BRFSS	South region	S. split	Sex	unknown	3–6	0.000	0.733	1.600
BRFSS	South region	T. Mondrian	Age	18_64	52985–53195	0.116	0.930	1.257
BRFSS	South region	T. Mondrian	Age	65_plus	30978–31185	0.242	0.844	1.480
BRFSS	South region	T. Mondrian	Age	unknown	1216–1249	0.127	0.944	1.398
BRFSS	South region	T. Mondrian	Sex	female	49770–49982	0.159	0.897	1.342
BRFSS	South region	T. Mondrian	Sex	male	35412–35627	0.166	0.902	1.338
BRFSS	South region	T. Mondrian	Sex	unknown	3–6	0.000	0.800	1.667
BRFSS	South region	T. split	Age	18_64	52985–53195	0.116	0.934	1.391
BRFSS	South region	T. split	Age	65_plus	30978–31185	0.242	0.839	1.580
BRFSS	South region	T. split	Age	unknown	1216–1249	0.127	0.951	1.602
BRFSS	South region	T. split	Sex	female	49770–49982	0.159	0.897	1.466
BRFSS	South region	T. split	Sex	male	35412–35627	0.166	0.904	1.459
BRFSS	South region	T. split	Sex	unknown	3–6	0.000	0.800	1.667
BRFSS	Later years	S. split	Age	18_64	104879–105148	0.096	0.957	1.326
BRFSS	Later years	S. split	Age	65_plus	59978–60210	0.216	0.901	1.576
BRFSS	Later years	S. split	Age	unknown	2819–2911	0.111	0.965	1.535
BRFSS	Later years	S. split	Sex	female	90389–90710	0.135	0.931	1.420
BRFSS	Later years	S. split	Sex	male	77272–77588	0.144	0.943	1.417
BRFSS	Later years	S. split	Sex	unknown	18–27	0.125	0.954	1.371
BRFSS	Later years	T. Mondrian	Age	18_64	104879–105148	0.096	0.934	1.286
BRFSS	Later years	T. Mondrian	Age	65_plus	59978–60210	0.216	0.840	1.487
BRFSS	Later years	T. Mondrian	Age	unknown	2819–2911	0.111	0.933	1.489
BRFSS	Later years	T. Mondrian	Sex	female	90389–90710	0.135	0.892	1.360
BRFSS	Later years	T. Mondrian	Sex	male	77272–77588	0.144	0.910	1.362
BRFSS	Later years	T. Mondrian	Sex	unknown	18–27	0.125	0.899	1.309
BRFSS	Later years	T. split	Age	18_64	104879–105148	0.096	0.932	1.223
BRFSS	Later years	T. split	Age	65_plus	59978–60210	0.216	0.842	1.416
BRFSS	Later years	T. split	Age	unknown	2819–2911	0.111	0.931	1.392
BRFSS	Later years	T. split	Sex	female	90389–90710	0.135	0.893	1.295
BRFSS	Later years	T. split	Sex	male	77272–77588	0.144	0.908	1.296
BRFSS	Later years	T. split	Sex	unknown	18–27	0.125	0.891	1.233
NHANES	Age 65+	S. split	Age	65_plus	3647–3647	0.279	0.374	1.101
NHANES	Age 65+	S. split	Sex	female	1844–1871	0.248	0.344	1.098
NHANES	Age 65+	S. split	Sex	male	1776–1803	0.311	0.404	1.104
NHANES	Age 65+	T. Mondrian	Age	65_plus	3647–3647	0.279	0.891	1.615
NHANES	Age 65+	T. Mondrian	Sex	female	1844–1871	0.248	0.894	1.619
NHANES	Age 65+	T. Mondrian	Sex	male	1776–1803	0.311	0.887	1.610

Data	Condition	Method	Variable	Subgroup	N	Prevalence	Coverage	Size
NHANES	Age 65+	T. split	Age	65_plus	3647–3647	0.279	0.894	1.806
NHANES	Age 65+	T. split	Sex	female	1844–1871	0.248	0.884	1.800
NHANES	Age 65+	T. split	Sex	male	1776–1803	0.311	0.904	1.812
NHANES	BMI threshold	S. split	Age	below 65 (all ages)	4997–5028	0.143	0.743	1.460
NHANES	BMI threshold	S. split	Age	65_plus	1218–1249	0.377	0.446	1.084
NHANES	BMI threshold	S. split	Sex	female	3498–3542	0.181	0.699	1.396
NHANES	BMI threshold	S. split	Sex	male	2704–2748	0.200	0.666	1.373
NHANES	BMI threshold	T. Mondrian	Age	below 65 (all ages)	4997–5028	0.143	0.937	1.378
NHANES	BMI threshold	T. Mondrian	Age	65_plus	1218–1249	0.377	0.753	1.502
NHANES	BMI threshold	T. Mondrian	Sex	female	3498–3542	0.181	0.907	1.398
NHANES	BMI threshold	T. Mondrian	Sex	male	2704–2748	0.200	0.893	1.408
NHANES	BMI threshold	T. split	Age	below 65 (all ages)	4997–5028	0.143	0.947	1.836
NHANES	BMI threshold	T. split	Age	65_plus	1218–1249	0.377	0.722	1.482
NHANES	BMI threshold	T. split	Sex	female	3498–3542	0.181	0.913	1.786
NHANES	BMI threshold	T. split	Sex	male	2704–2748	0.200	0.890	1.740
NHANES	Race proxy	S. split	Age	below 65 (all ages)	5345–5366	0.064	0.965	1.167
NHANES	Race proxy	S. split	Age	65_plus	723–744	0.346	0.806	1.563
NHANES	Race proxy	S. split	Sex	female	3036–3096	0.100	0.946	1.206
NHANES	Race proxy	S. split	Sex	male	2993–3053	0.096	0.946	1.225
NHANES	Race proxy	T. Mondrian	Age	below 65 (all ages)	5345–5366	0.064	0.937	1.173
NHANES	Race proxy	T. Mondrian	Age	65_plus	723–744	0.346	0.627	1.335
NHANES	Race proxy	T. Mondrian	Sex	female	3036–3096	0.100	0.906	1.187
NHANES	Race proxy	T. Mondrian	Sex	male	2993–3053	0.096	0.892	1.197
NHANES	Race proxy	T. split	Age	below 65 (all ages)	5345–5366	0.064	0.933	1.069
NHANES	Race proxy	T. split	Age	65_plus	723–744	0.346	0.648	1.233
NHANES	Race proxy	T. split	Sex	female	3036–3096	0.100	0.900	1.086
NHANES	Race proxy	T. split	Sex	male	2993–3053	0.096	0.896	1.093
NHANES	5-year years	S. split	Age	below 65 (all ages)	10744–10791	0.059	0.955	1.252
NHANES	5-year years	S. split	Age	65_plus	1903–1950	0.304	0.702	1.452
NHANES	5-year years	S. split	Sex	female	6425–6486	0.087	0.911	1.279
NHANES	5-year years	S. split	Sex	male	6208–6269	0.106	0.923	1.287
NHANES	5-year years	T. Mondrian	Age	below 65 (all ages)	10744–10791	0.059	0.945	1.156
NHANES	5-year years	T. Mondrian	Age	65_plus	1903–1950	0.304	0.655	1.368
NHANES	5-year years	T. Mondrian	Sex	female	6425–6486	0.087	0.895	1.183
NHANES	5-year years	T. Mondrian	Sex	male	6208–6269	0.106	0.908	1.194

Data	Condition	Method	Variable	Subgroup	N	Prevalence	Coverage	Size
NHANES	5 later years	T. split	Age	below 65 (all ages)	10744–10791	0.059	0.946	1.220
NHANES	5 later years	T. split	Age	65_plus	1903–1950	0.304	0.652	1.360
NHANES	5 later years	T. split	Sex	female	6425–6486	0.087	0.895	1.237
NHANES	5 later years	T. split	Sex	male	6208–6269	0.106	0.908	1.246
UCI	Age 70+	S. split	Age	older70	22188–22188	0.101	0.803	1.555
UCI	Age 70+	S. split	Race	AfricanAmerican	2724–2747	0.098	0.808	1.554
UCI	Age 70+	S. split	Race	Asian	152–168	0.127	0.910	1.527
UCI	Age 70+	S. split	Race	Caucasian	18109–18155	0.102	0.797	1.555
UCI	Age 70+	S. split	Race	Hispanic	263–294	0.121	0.845	1.575
UCI	Age 70+	S. split	Race	Other	256–283	0.085	0.896	1.553
UCI	Age 70+	S. split	Race	missing	594–612	0.086	0.868	1.557
UCI	Age 70+	T. Mondrian	Age	older70	22188–22188	0.101	0.903	1.722
UCI	Age 70+	T. Mondrian	Race	AfricanAmerican	2724–2747	0.098	0.907	1.736
UCI	Age 70+	T. Mondrian	Race	Asian	152–168	0.127	0.948	1.532
UCI	Age 70+	T. Mondrian	Race	Caucasian	18109–18155	0.102	0.900	1.727
UCI	Age 70+	T. Mondrian	Race	Hispanic	263–294	0.121	0.918	1.709
UCI	Age 70+	T. Mondrian	Race	Other	256–283	0.085	0.943	1.579
UCI	Age 70+	T. Mondrian	Race	missing	594–612	0.086	0.934	1.617
UCI	Age 70+	T. split	Age	older70	22188–22188	0.101	0.903	1.749
UCI	Age 70+	T. split	Race	AfricanAmerican	2724–2747	0.098	0.906	1.751
UCI	Age 70+	T. split	Race	Asian	152–168	0.127	0.954	1.695
UCI	Age 70+	T. split	Race	Caucasian	18109–18155	0.102	0.900	1.750
UCI	Age 70+	T. split	Race	Hispanic	263–294	0.121	0.922	1.742
UCI	Age 70+	T. split	Race	Other	256–283	0.085	0.949	1.728
UCI	Age 70+	T. split	Race	missing	594–612	0.086	0.946	1.745
UCI	Admission source	S. split	Age	older70	12245–12337	0.106	0.824	1.469
UCI	Admission source	S. split	Age	younger	14465–14557	0.078	0.889	1.509
UCI	Admission source	S. split	Race	AfricanAmerican	5372–5449	0.083	0.874	1.502
UCI	Admission source	S. split	Race	Asian	227–238	0.084	0.850	1.486
UCI	Admission source	S. split	Race	Caucasian	19445–19529	0.095	0.853	1.487
UCI	Admission source	S. split	Race	Hispanic	673–691	0.073	0.872	1.507
UCI	Admission source	S. split	Race	Other	484–524	0.068	0.899	1.510
UCI	Admission source	S. split	Race	missing	493–521	0.071	0.869	1.491
UCI	Admission source	T. Mondrian	Age	older70	12245–12337	0.106	0.880	1.719
UCI	Admission source	T. Mondrian	Age	younger	14465–14557	0.078	0.919	1.692
UCI	Admission source	T. Mondrian	Race	AfricanAmerican	5372–5449	0.083	0.914	1.730
UCI	Admission source	T. Mondrian	Race	Asian	227–238	0.084	0.885	1.728
UCI	Admission source	T. Mondrian	Race	Caucasian	19445–19529	0.095	0.897	1.701

Data	Condition	Method	Variable	Subgroup	N	Prevalence	Coverage	Size
UCI	Admission source	T. Mondrian	Race	Hispanic	673–691	0.073	0.895	1.729
UCI	Admission source	T. Mondrian	Race	Other	484–524	0.068	0.925	1.589
UCI	Admission source	T. Mondrian	Race	missing	493–521	0.071	0.914	1.658
UCI	Admission source	T. split	Age	older70	12245–12337	0.106	0.879	1.558
UCI	Admission source	T. split	Age	younger	14465–14557	0.078	0.919	1.571
UCI	Admission source	T. split	Race	AfricanAmerican	5372–5449	0.083	0.909	1.566
UCI	Admission source	T. split	Race	Asian	227–238	0.084	0.887	1.567
UCI	Admission source	T. split	Race	Caucasian	19445–19529	0.095	0.897	1.564
UCI	Admission source	T. split	Race	Hispanic	673–691	0.073	0.908	1.583
UCI	Admission source	T. split	Race	Other	484–524	0.068	0.927	1.560
UCI	Admission source	T. split	Race	missing	493–521	0.071	0.917	1.567
UCI	Race proxy	S. split	Age	older70	2736–2767	0.100	0.880	1.590
UCI	Race proxy	S. split	Age	younger	6254–6285	0.079	0.924	1.578
UCI	Race proxy	S. split	Race	AfricanAmerican	9021–9021	0.085	0.911	1.582
UCI	Race proxy	T. Mondrian	Age	older70	2736–2767	0.100	0.862	1.706
UCI	Race proxy	T. Mondrian	Age	younger	6254–6285	0.079	0.917	1.703
UCI	Race proxy	T. Mondrian	Race	AfricanAmerican	9021–9021	0.085	0.900	1.704
UCI	Race proxy	T. split	Age	older70	2736–2767	0.100	0.860	1.554
UCI	Race proxy	T. split	Age	younger	6254–6285	0.079	0.915	1.539
UCI	Race proxy	T. split	Race	AfricanAmerican	9021–9021	0.085	0.898	1.543

S3. Finite-rank analytical examples

For continuous independent identically distributed scores, expected coverage is $k_n/(n+1)$, or one if the rank overflows. For $1 \leq k_n \leq n$, uniform-score order statistics have standard deviation

$$\sqrt{k_n(n+1-k_n)/[(n+1)^2(n+2)]}.$$

Under rank overflow, the threshold is $+\infty$; its ordinary real-valued standard deviation is undefined, so the table reports N/A. Coverage remains one for finite future scores. These are analytical examples, not health-data experiments.

S4. Full outcome-model inputs

The UCI prediction time is completion of the index hospitalization. Inputs include discharge disposition and other completed-stay information; this is a retrospective readmission task, not an admission-time prediction model.

Table S3: Finite-rank examples at 90% coverage.

n	k_n	Expected coverage	Threshold SD
8	9	1.0000	N/A
9	9	0.9000	0.0905
26	25	0.9259	0.0495
44	41	0.9111	0.0420
91	83	0.9022	0.0308
300	271	0.9003	0.0172

brfss_multiyear_diabetes Numeric outcome features: `physhlth_days`, `menthlth_days`, `bmi`, `age80`.

Categorical outcome features: `survey_year`, `_STATE`, `region`, `GENHLTH`, `HLTHPLN1`, `PERSDOC2`, `MEDCOST`, `CHECKUP1`, `BPHIGH4`, `BLOODCHO`, `CHOLCHK`, `TOLDHI2`, `CVDINFR4`, `CVDCRHD4`, `CVDSTRK3`, `ASTHMA3`, `CHCSCNCR`, `CHCOCNCR`, `CHCCOPD1`, `HAVARTH3`, `ADDEPEV2`, `CHCKIDNY`, `SEX`, `MARITAL`, `EDUCA`, `EMPLOY1`, `INCOME2`, `QLACTLM2`, `BLIND`, `DECIDE`, `DIFFWALK`, `SMOKE100`, `SMOKDAY2`, `ALCDAY5`, `EXERANY2`, `_RACE`, `_AGEG5YR`, `_AGE65YR`, `_BMI5CAT`, `_RFHYPE5`, `_RFCHOL`, `_MICHD`, `_SMOKER3`, `_RFSMOK3`, `_TOTINDA`, `_FRTLT1`, `_VEGLT1`, `_PA150R2`.

nhanes_multicycle_diabetes Numeric outcome features: `RIDAGEYR`, `INDFMPIR`, `BMXBMI`, `BMXWT`, `BMXHT`, `BMXWAIST`.

Categorical outcome features: `cycle_start`, `RIAGENDR`, `RIDRETH3`, `DMDEDUC2`, `DMDMARTL`, `BPQ020`, `BPQ080`, `SMQ020`, `MCQ160B`, `MCQ160C`, `MCQ160F`, `MCQ160M`.

uci_diabetes_readmission Numeric outcome features: `age_mid`, `time_in_hospital`, `num_lab_procedures`, `num_procedures`, `num_medications`, `number_outpatient`, `number_emergency`, `number_inpatient`, `number_diagnoses`.

Categorical outcome features: `race`, `gender`, `admission_type_id`, `discharge_disposition_id`, `admission_source_id`, `admission_mode`, `max_glu_serum`, `A1Cresult`, `change`, `diabetesMed`, `diag_1_group`, `diag_2_group`, `diag_3_group`, `metformin`, `repaglinide`, `nateglinide`, `chlorpropamide`, `glimepiride`, `acetohexamide`, `glipizide`, `glyburide`, `tolbutamide`, `pioglitazone`, `rosiglitazone`, `acarbose`, `miglitol`, `troglitazone`, `tolazamide`, `examide`, `citoglipton`, `insulin`, `glyburide-metformin`, `glipizide-metformin`, `glimepiride-pioglitazone`, `metformin-rosiglitazone`, `metformin-pioglitazone`.

S5. Adult NHANES sensitivity

Age must be at least 18; the BMI condition requires observed BMI. All four conditions and five source-model seeds are refitted, with five draws of 300 source and 300 target labels. This averages four NHANES conditions, not the three-dataset benchmark.

S6. Repeated-seed and calibration-draw dispersion

Table S5: Adult NHANES results.

Method	Coverage	Size	Class gap
Source split	0.7627	1.3464	0.2888
Source Mondrian	0.7656	1.3623	0.2898
Target split	0.8994	1.5690	0.1211
Target Mondrian	0.9010	1.4813	0.0367

Table S6: Seed and draw dispersion of additional outcome-blind 10% coverage.

Condition	Method	Seed min	Seed max	Seed SD	Draw SD
BRFSS Age 65+	Source Mondrian	0.7815	0.8288	0.0193	0.0000
BRFSS Age 65+	Source split	0.7780	0.8253	0.0203	0.0000
BRFSS Age 65+	Target Mondrian	0.8977	0.9055	0.0033	0.0020
BRFSS Age 65+	Target split	0.8972	0.9041	0.0032	0.0020
BRFSS BMI threshold	Source Mondrian	0.6457	0.7469	0.0413	0.0000
BRFSS BMI threshold	Source split	0.6392	0.7471	0.0435	0.0000
BRFSS BMI threshold	Target Mondrian	0.8964	0.9015	0.0018	0.0021
BRFSS BMI threshold	Target split	0.8960	0.9007	0.0018	0.0020
BRFSS Race proxy	Source Mondrian	0.8597	0.8709	0.0050	0.0000
BRFSS Race proxy	Source split	0.8578	0.8706	0.0058	0.0000
BRFSS Race proxy	Target Mondrian	0.8964	0.9016	0.0022	0.0042
BRFSS Race proxy	Target split	0.8982	0.9003	0.0009	0.0035
BRFSS South region	Source Mondrian	0.8637	0.8721	0.0038	0.0000
BRFSS South region	Source split	0.8637	0.8717	0.0037	0.0000
BRFSS South region	Target Mondrian	0.8970	0.9021	0.0019	0.0029
BRFSS South region	Target split	0.8985	0.9020	0.0014	0.0026
BRFSS Later years	Source Mondrian	0.9224	0.9402	0.0073	0.0000
BRFSS Later years	Source split	0.9218	0.9552	0.0146	0.0000
BRFSS Later years	Target Mondrian	0.8985	0.9018	0.0013	0.0012
BRFSS Later years	Target split	0.8977	0.9018	0.0015	0.0010
NHANES Age 65+	Source Mondrian	0.3279	0.4590	0.0545	0.0000
NHANES Age 65+	Source split	0.3274	0.4565	0.0545	0.0000
NHANES Age 65+	Target Mondrian	0.8814	0.9016	0.0075	0.0117
NHANES Age 65+	Target split	0.8877	0.9032	0.0064	0.0132
NHANES BMI threshold	Source Mondrian	0.5680	0.7653	0.0813	0.0000
NHANES BMI threshold	Source split	0.5634	0.7642	0.0833	0.0000
NHANES BMI threshold	Target Mondrian	0.8967	0.9100	0.0054	0.0075
NHANES BMI threshold	Target split	0.8970	0.9100	0.0047	0.0084
NHANES Race proxy	Source Mondrian	0.9225	0.9534	0.0146	0.0000
NHANES Race proxy	Source split	0.9235	0.9821	0.0231	0.0000
NHANES Race proxy	Target Mondrian	0.8867	0.9130	0.0096	0.0085
NHANES Race proxy	Target split	0.8847	0.9080	0.0117	0.0057
NHANES Later years	Source Mondrian	0.8861	0.9234	0.0155	0.0000
NHANES Later years	Source split	0.8862	0.9319	0.0178	0.0000
NHANES Later years	Target Mondrian	0.8919	0.9059	0.0057	0.0049
NHANES Later years	Target split	0.8920	0.9062	0.0056	0.0043
UCI Age 70+	Source Mondrian	0.7495	0.8202	0.0329	0.0000
UCI Age 70+	Source split	0.7498	0.8246	0.0301	0.0000

Table S6: Seed and draw dispersion of additional outcome-blind 10% coverage.

Condition	Method	Seed min	Seed max	Seed SD	Draw SD
UCI Age 70+	Target Mondrian	0.9005	0.9059	0.0020	0.0036
UCI Age 70+	Target split	0.8995	0.9062	0.0026	0.0038
UCI Admission source	Source Mondrian	0.8294	0.8758	0.0186	0.0000
UCI Admission source	Source split	0.8245	0.8817	0.0235	0.0000
UCI Admission source	Target Mondrian	0.8981	0.9058	0.0039	0.0037
UCI Admission source	Target split	0.8942	0.9044	0.0042	0.0039
UCI Race proxy	Source Mondrian	0.8874	0.9309	0.0154	0.0000
UCI Race proxy	Source split	0.8937	0.9383	0.0169	0.0000
UCI Race proxy	Target Mondrian	0.8956	0.9059	0.0044	0.0070
UCI Race proxy	Target split	0.8928	0.9051	0.0056	0.0077

These ranges and standard deviations describe specified random partitions and draws. They are not confidence intervals over independent deployments. Reference CSVs retain every condition, seed, draw, budget, and method.

S7. Optimizer sensitivity and independently solved controls

The sensitivity design includes all twelve conditions at seed 0 and all five model seeds for NHANES age 65+, with five draws: 80 problems per method. The age columns average its 25 seed–draw problems; objective and gradient columns summarize all 80 problems. OT gradient is with respect to softmax logits. LR-QR gradient is the projected gradient at the lowest-objective start, with the beta component respecting its bounds. The pinball objective is nonsmooth; these residuals are diagnostics rather than certificates. Uniform initialization is used for OT budgets; perturbed logits use normal noise with standard deviation 0.2 and the displayed seeds. LR-QR standard starts are 0.25, 1, 4, and extra starts are 0.1, 2, 10; the increased CV budget is shared by both final-fit start sets. The full main comparison uses 4000 OT and 1800 CV/8000 final LR-QR steps on 300 draws, separate from this subset. Every budget and initialization remains in the reference results.

Table S7: Optimizer sensitivity: age-condition outcomes and 80-problem objective/gradient diagnostics.

Method	Budget/initialization	Age class min	Age coverage	Age two-label	Objective	Gradient mean	Gradient max
OT	1,000 steps	0.4203	0.5820	0.3642	0.4057	0.0110	0.0187
OT	4,000; jitter 731	0.9064	0.9325	0.8747	0.2858	0.0042	0.0148
OT	4,000; jitter 991	0.9057	0.9319	0.8737	0.2859	0.0042	0.0143
OT	4,000 steps	0.9076	0.9333	0.8762	0.2858	0.0042	0.0147
OT	8,000 steps	0.9078	0.9335	0.8765	0.2745	0.0008	0.0019
LR-QR	600 CV / 2,000 final	0.7197	0.7940	0.7043	-0.9183	0.1409	0.3970
LR-QR	1,800 / 8,000; extra starts	0.6088	0.7135	0.5942	-0.9196	0.1333	0.2664
LR-QR	1,800 CV / 8,000 final	0.6074	0.7125	0.5927	-0.9196	0.1307	0.2921

For OT controls, 300 calibration scores lie on two repeated support points; 300 target probabilities define the same auxiliary distributions and grid as the health implementation. The objective depends only on total mass at one support point. Between CDF breakpoints it is quadratic, so enumerating breakpoints, interval stationary points and endpoints yields its global minimum independently of gradient optimization. We solve three controls and compare 1000, 4000 and 8000 steps.

For LR-QR controls, all covariate vectors are an intercept and 300 scores are equally spaced from 0.05 to 0.95. The objective is $L(h) + \lambda(\beta h - 1)^2 - \lambda$. Writing $q = s_{(270)}$, the lower bound $L(q) - \lambda$ is attained at $h = q$ and $\beta = 1/q$, within the beta bounds. For the discrete pinball loss any h between the 270th and 271st scores minimizes L; the reference selects the lower endpoint. We compare three lambda values (0.1, 0.4, 0.8), three starts (0.25, 1, 4), and 2000/8000 steps. The independently calculated objective agrees with the fitted implementation within 10^{-12}. These control minima certify only these constructed problems.

Table S8: Tractable controls: maximum gaps to independently solved global objectives.

Method	Steps	Controls	Max global gap	Objective agreement
OT	1000	3	2.13e-04	1.73e-07
OT	4000	3	6.05e-13	1.73e-07
OT	8000	3	5.89e-12	6.79e-07
LR-QR	2000	9	4.34e-06	$\leq 10^{-12}$
LR-QR	8000	9	3.50e-06	$\leq 10^{-12}$

Independent evaluation of the fitted OT objective has maximum discrepancy 1.55×10^{-7} from its float32 evaluation. This agreement concerns the same fitted weights, distinct from the global-control optimization gaps and budget sensitivity reported above. Reconstruction procedures and computational measurement counts are documented in the supporting materials.

S8. Fixed-predictor discrimination and probability accuracy

AUROC, average precision (AP) and Brier score use the unchanged source-fitted elastic-net predictions on each untouched target test role. These metrics average five model seeds within condition and involve no recalibration, method selection or tuning on test outcomes. Counts are identical across seeds for these roles. Prevalence is reported because AP depends on outcome frequency. Brier score is the mean squared error between the predicted probability and binary outcome. Its magnitude depends on outcome prevalence and does not by itself establish probability calibration. Near-nominal conformal coverage does not imply a low Brier score.

Table S9: Fixed-predictor target-test diagnostics.

Condition	N test	Prevalence	AUROC	AP	Brier
BRFSS Age 65+	101,741	0.2128	0.7519	0.4363	0.2604
BRFSS BMI threshold	82,468	0.2376	0.7879	0.4974	0.3375
BRFSS Race proxy	19,195	0.2094	0.8179	0.5095	0.1998
BRFSS South region	85,400	0.1619	0.8267	0.4587	0.1968
BRFSS Later years	168,000	0.1392	0.8197	0.4096	0.1496
NHANES Age 65+	3,647	0.2790	0.6682	0.4199	0.5533
NHANES BMI threshold	6,246	0.1894	0.8000	0.4529	0.3791
NHANES Race proxy	6,089	0.0980	0.8834	0.4033	0.0948
NHANES Later years	12,694	0.0964	0.8841	0.4122	0.1401
UCI Age 70+	22,188	0.1014	0.6163	0.1562	0.2966
UCI Admission source	26,802	0.0907	0.6299	0.1489	0.2659
UCI Race proxy	9,021	0.0850	0.6233	0.1473	0.2399

S9. Predictor-family sensitivity

LightGBM 4.6.0 uses 120 trees, 31 leaves, learning rate 0.05, balanced class weights, random state equal to the model seed and two CPU threads; remaining parameters are defaults. Source-fitted preprocessing is the same as in the primary model. LightGBM uses the additional outcome-blind role allocations and five 10% target draws per condition–seed. Source and target thresholds use Supplementary Equation S1, with exact finite ranks and no empty-set completion.

Table S10: Nonlinear predictor sensitivity.

Predictor	Method	Coverage	Size	Class gap
Elastic-net	Source split	0.8066	1.3937	0.2576
Elastic-net	Source Mondrian	0.8050	1.4011	0.2074
Elastic-net	Target split	0.8998	1.5416	0.1472
Elastic-net	Target Mondrian	0.8997	1.4865	0.0161
LightGBM	Source split	0.8314	1.3669	0.1697
LightGBM	Source Mondrian	0.8297	1.4074	0.1521
LightGBM	Target split	0.9002	1.4989	0.1025
LightGBM	Target Mondrian	0.8999	1.4789	0.0146

S10. Feature-ablation and L2 sensitivity analyses

The feature ablation drops the condition-defining variables listed below, then refits every condition and seed. The sensitivity fits use the additional outcome-blind protocol: its role allocations and labels, the exact thresholds in Supplementary Equation S1, and five 10% target draws per condition–seed. BRFSS drops survey year, region/state, age encodings, BMI encodings, or race; NHANES drops cycle, age, all four body-size variables, or race; UCI drops age, race, or admission-source/mode as appropriate. Exact lists are in the supplied protocol. The L2 comparison keeps all features and changes only the logistic penalty. Each table retains all twelve conditions. No condition or fit was selected using test results.

Table S11: Shift-defining feature ablation.

Condition	Source cov.	Source shortfall	Target cov.	Target shortfall	Target size
BRFSS Age 65+	0.7632	0.1368	0.9003	0.0011	1.7065
BRFSS BMI threshold	0.8658	0.0342	0.8991	0.0017	1.5540
BRFSS Race proxy	0.8734	0.0266	0.8990	0.0019	1.4856
BRFSS South region	0.8648	0.0352	0.8998	0.0012	1.4688
BRFSS Later years	0.9243	0.0000	0.9001	0.0005	1.2999
NHANES Age 65+	0.6077	0.2923	0.8945	0.0079	1.7991
NHANES BMI threshold	0.8437	0.0563	0.9034	0.0030	1.7319
NHANES Race proxy	0.9306	0.0000	0.8986	0.0048	1.2230
NHANES Later years	0.8852	0.0148	0.9006	0.0018	1.2456
UCI Age 70+	0.8124	0.0876	0.9024	0.0009	1.6586
UCI Admission source	0.8758	0.0242	0.8995	0.0022	1.4853
UCI Race proxy	0.9125	0.0000	0.9001	0.0033	1.5195

Table S12: L2 logistic SGD sensitivity.

Condition	Source cov.	Source shortfall	Target cov.	Target shortfall	Target size
BRFSS Age 65+	0.8122	0.0878	0.9005	0.0014	1.6568
BRFSS BMI threshold	0.6775	0.2225	0.8986	0.0020	1.7639
BRFSS Race proxy	0.8545	0.0455	0.8981	0.0028	1.5397
BRFSS South region	0.8653	0.0347	0.8997	0.0016	1.4668
BRFSS Later years	0.9423	0.0000	0.9001	0.0006	1.2147
NHANES Age 65+	0.3749	0.5251	0.8945	0.0081	1.8091
NHANES BMI threshold	0.7281	0.1719	0.9019	0.0029	1.6988
NHANES Race proxy	0.9510	0.0000	0.8977	0.0055	1.0978
NHANES Later years	0.9266	0.0033	0.9002	0.0020	1.2238
UCI Age 70+	0.8507	0.0493	0.9020	0.0011	1.6238
UCI Admission source	0.8785	0.0215	0.8995	0.0024	1.6835
UCI Race proxy	0.9019	0.0043	0.8997	0.0037	1.5008

S11. Random controls and clipped-weighting comparison

The whole-dataset random controls use a 25% test fraction and a 30% calibration fraction within development rows and five seeds per dataset. They are not matched to each condition's source and target sample sizes or prevalences. The source-trained model and exact rank rule are the same as in the additional outcome-blind workflow. Fifteen controls are not sixty independent deployment controls. The clipped-domain-weighting baseline uses the stated empirical weighted-CDF threshold

Table S13: Whole-dataset random controls.

Data	Seed	Train N	Cal. N	Test N	Coverage	Size
BRFSS	0	220500	94500	105000	0.9017	1.4049
BRFSS	1	220500	94500	105000	0.8999	1.3974
BRFSS	2	220500	94500	105000	0.9000	1.3584
BRFSS	3	220500	94500	105000	0.8986	1.3679
BRFSS	4	220500	94500	105000	0.9002	1.4718
NHANES	0	19394	8313	9236	0.8937	1.2348
NHANES	1	19394	8313	9236	0.8978	1.2814
NHANES	2	19394	8313	9236	0.9021	1.2162
NHANES	3	19394	8313	9236	0.9042	1.2866
NHANES	4	19394	8313	9236	0.8969	1.1887
UCI	0	37545	16091	17879	0.9039	1.7395
UCI	1	37545	16091	17879	0.8944	1.3241
UCI	2	37545	16091	17879	0.9003	1.8603
UCI	3	37545	16091	17879	0.8986	1.8608
UCI	4	37545	16091	17879	0.9014	1.7228

rule and full source calibration sample. Each seed receives one unlabeled 10% target draw, without test features or target outcomes. Empty sets are permitted. This threshold is not assigned a distribution-free weighted validity guarantee.

S12. Score-threshold diagnostic

The source threshold is fitted on source calibration labels. The target10 threshold is the mean across five labeled target-pool draws and five model seeds. The empirical target-test percentile

Table S14: Clipped weighting: condition means.

Condition	Coverage	Size	Empty	Singleton	Two-label	Class gap
BRFSS Age 65+	0.8094	1.4486	0.0000	0.5514	0.4486	0.1563
BRFSS BMI threshold	0.6818	1.3314	0.0000	0.6686	0.3314	0.4017
BRFSS Race proxy	0.8707	1.4037	0.0000	0.5963	0.4037	0.0960
BRFSS South region	0.8681	1.3779	0.0000	0.6221	0.3779	0.0886
BRFSS Later years	0.9367	1.4187	0.0000	0.5813	0.4187	0.0552
NHANES Age 65+	0.7570	1.5955	0.0000	0.4045	0.5955	0.3371
NHANES BMI threshold	0.8571	1.6855	0.0000	0.3145	0.6855	0.1698
NHANES Race proxy	0.9458	1.2151	0.0000	0.7849	0.2151	0.1051
NHANES Later years	0.9168	1.2822	0.0000	0.7178	0.2822	0.0574
UCI Age 70+	0.8084	1.5647	0.0000	0.4353	0.5647	0.1923
UCI Admission source	0.8591	1.4909	0.0000	0.5091	0.4909	0.3599
UCI Race proxy	0.9105	1.5814	0.0000	0.4186	0.5814	0.2322

is retrospective, uses linear interpolation, and never fits a deployable threshold. It shares score information with source coverage shortfall. Their correlation is descriptive, not independent causal evidence.

Table S15: Threshold diagnostics: condition means.

Condition	Source q	Test $q_{.9}$	Target10 q	Source shortfall
BRFSS Age 65+	0.7107	0.8023	0.8028	0.0907
BRFSS BMI threshold	0.7174	0.9106	0.9099	0.2196
BRFSS Race proxy	0.7162	0.7645	0.7630	0.0352
BRFSS South region	0.7265	0.7723	0.7719	0.0319
BRFSS Later years	0.7251	0.6590	0.6592	0.0000
NHANES Age 65+	0.7194	0.9724	0.9708	0.5263
NHANES BMI threshold	0.7697	0.9281	0.9299	0.2153
NHANES Race proxy	0.7344	0.5982	0.5965	0.0000
NHANES Later years	0.7613	0.7177	0.7214	0.0028
UCI Age 70+	0.6751	0.7541	0.7577	0.0971
UCI Admission source	0.6701	0.7188	0.7179	0.0409
UCI Race proxy	0.6676	0.6555	0.6514	0.0013

S13. Target-label observation mechanisms and protocol choice

The source predictors, pools and test roles are fixed. At 100 or 300 calibration labels, we compare uniform observation, positive-outcome depletion and risk-prioritized review. Relative weights and Bernoulli normalization are specified in Supplementary Section S20.3 and the executable protocol. Eligibility is not redrawn to meet a quota; all observed quotas meet their planned count. The sentinel is an alternative uniform labeling protocol, independent of the available-label observation mechanism. Its label count is not a claim about combined or monetary acquisition cost.

The protocol rule uses sampling-channel metadata only. It chooses available Mondrian under uniform/outcome-only observation and sentinel Mondrian under risk-prioritized observation. This is

a known-design comparison, not an estimated clinical missingness classifier or an optimal allocation method. Means average draws, seeds, conditions and then three datasets equally. Tables S16–S17 summarize both budgets and all five methods and their class counts. Condition-specific metrics appear in the supporting file `results/label_acquisition/condition_summary.csv`; the full reference CSVs retain every draw and both class coverages.

Table S16: Complete label-observation comparison: equal-dataset means. C_0 and C_1 are outcome-class prediction-set coverage.

Labels	Channel	Method	Coverage	C_0	C_1	Size	Correct single	Incorrect single
100	Outcome-dependent	Available Mondrian	0.9176	0.9047	0.9952	1.8567	0.0608	0.0824
100	Outcome-dependent	Available split	0.9023	0.9008	0.8671	1.5475	0.3512	0.0940
100	Outcome-dependent	Protocol rule	0.9176	0.9047	0.9952	1.8567	0.0608	0.0824
100	Outcome-dependent	Sentinel Mondrian	0.9090	0.9048	0.9363	1.6181	0.2910	0.0910
100	Outcome-dependent	Sentinel split	0.9010	0.8960	0.8857	1.5499	0.3502	0.0982
100	Risk-prioritized	Available Mondrian	0.9347	0.9456	0.8736	1.5285	0.4062	0.0653
100	Risk-prioritized	Available split	0.9343	0.9308	0.9196	1.6418	0.2925	0.0657
100	Risk-prioritized	Protocol rule	0.9066	0.9028	0.9348	1.6105	0.2962	0.0934
100	Risk-prioritized	Sentinel Mondrian	0.9066	0.9028	0.9348	1.6105	0.2962	0.0934
100	Risk-prioritized	Sentinel split	0.8995	0.8946	0.8817	1.5432	0.3559	0.1001
100	Uniform	Available Mondrian	0.9076	0.9031	0.9376	1.6221	0.2855	0.0924
100	Uniform	Available split	0.9000	0.8954	0.8798	1.5459	0.3530	0.0989
100	Uniform	Protocol rule	0.9076	0.9031	0.9376	1.6221	0.2855	0.0924
100	Uniform	Sentinel Mondrian	0.9090	0.9060	0.9332	1.6049	0.3038	0.0907
100	Uniform	Sentinel split	0.9011	0.8963	0.8846	1.5496	0.3504	0.0978
300	Outcome-dependent	Available Mondrian	0.9088	0.9015	0.9611	1.6966	0.2122	0.0912
300	Outcome-dependent	Available split	0.9020	0.9000	0.8674	1.5419	0.3585	0.0965
300	Outcome-dependent	Protocol rule	0.9088	0.9015	0.9611	1.6966	0.2122	0.0912
300	Outcome-dependent	Sentinel Mondrian	0.9021	0.9011	0.9109	1.5125	0.3896	0.0979
300	Outcome-dependent	Sentinel split	0.9008	0.8963	0.8815	1.5458	0.3549	0.0992
300	Risk-prioritized	Available Mondrian	0.9318	0.9449	0.8527	1.4871	0.4447	0.0682
300	Risk-prioritized	Available split	0.9359	0.9322	0.9222	1.6454	0.2905	0.0641
300	Risk-prioritized	Protocol rule	0.9023	0.9013	0.9092	1.5124	0.3899	0.0977
300	Risk-prioritized	Sentinel Mondrian	0.9023	0.9013	0.9092	1.5124	0.3899	0.0977
300	Risk-prioritized	Sentinel split	0.9003	0.8957	0.8813	1.5442	0.3561	0.0997
300	Uniform	Available Mondrian	0.9029	0.9014	0.9122	1.5161	0.3868	0.0971
300	Uniform	Available split	0.9000	0.8951	0.8822	1.5441	0.3557	0.0998
300	Uniform	Protocol rule	0.9029	0.9014	0.9122	1.5161	0.3868	0.0971
300	Uniform	Sentinel Mondrian	0.9023	0.9023	0.9043	1.5037	0.3986	0.0977
300	Uniform	Sentinel split	0.9004	0.8959	0.8802	1.5449	0.3553	0.0994

S14. Independent institutional evaluation

The four cardiac centers contain 920 diagnostic records; each target center is evaluated against the other three pooled as source. The outcome is contemporaneous diagnosis, not future disease incidence. The sepsis cohort comprises 24,095 eligible patients (10,174 from hospital A and 13,921 from B), with prediction at the end of hour 12 and supplied-label transition in hours 13–24. Table S23 gives full source/target and calibration class counts. Both hospital directions are evaluated. The institutional collection means remain separate from the twelve primary conditions.

The external collections are independent of BRFSS, NHANES and UCI readmission; the six comparisons within them are related. The protocol fixes four leave-one-center-out cardiac conditions and both sepsis hospital directions, two predictors, five source-model seeds and five target draws before inspecting external coverage. All six conditions, eight label fractions and four calibration workflows contribute to the reported summaries; quota-shortfall draws use all eligible labels. Supplementary tables retain every condition, both outcome classes and set size. The complete eight-fraction and observation-channel reference summaries and executable regeneration and independent verification

Table S17: Calibration and class counts. Ranges span all 300 condition–seed–draw combinations per row. Positive-class overflow counts draws with fewer than nine positive labels; Mondrian then always includes that candidate label.

Labels	Channel	Design	N used	Class 0	Class 1	Class-1 overflow
100	Outcome-dependent	available	100–100	89–100	0–11	286
100	Outcome-dependent	sentinel	100–100	62–97	3–38	58
100	Risk-prioritized	available	100–100	57–97	3–43	21
100	Risk-prioritized	sentinel	100–100	64–99	1–36	52
100	Uniform	available	100–100	61–98	2–39	54
100	Uniform	sentinel	100–100	60–98	2–40	51
300	Outcome-dependent	available	300–300	270–300	0–30	124
300	Outcome-dependent	sentinel	300–300	201–288	12–99	0
300	Risk-prioritized	available	300–300	193–282	18–107	0
300	Risk-prioritized	sentinel	300–300	203–283	17–97	0
300	Uniform	available	300–300	195–288	12–105	0
300	Uniform	sentinel	300–300	201–286	14–99	0

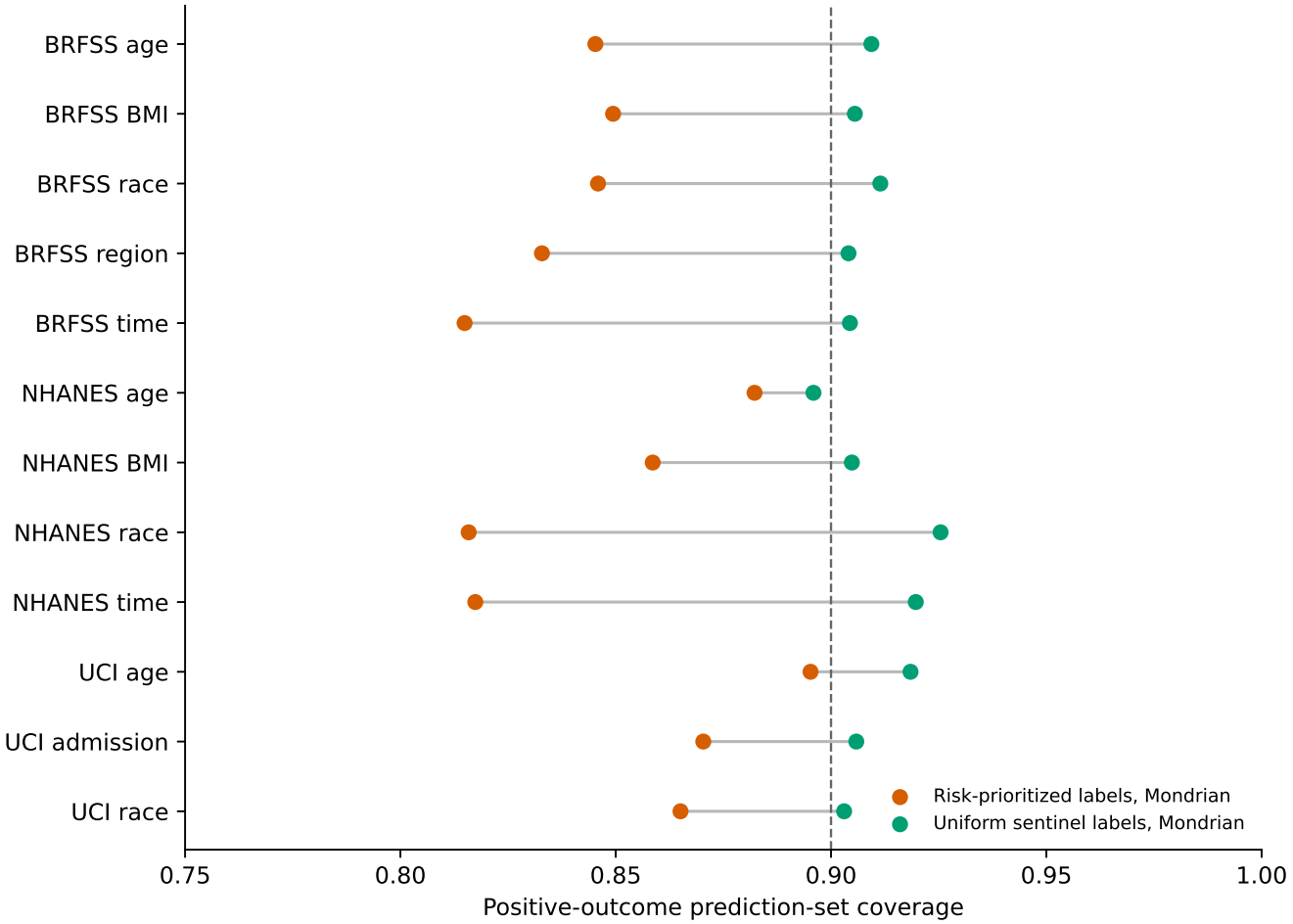


Figure S1: Positive-outcome coverage with 300 calibration labels under risk-prioritized observation, for every condition. Points average five source-model seeds and five calibration draws. The horizontal range is 0.75–1.00; the dashed reference is 0.90. Uniform sentinel labels are a counterfactual calibration-data alternative, not a causal estimate of a real clinical intervention.

programs are supplied in Supplementary Code 1.

The cardiac files are `processed.cleveland.data`, `processed.hungarian.data`, `processed.switzerland.data` and `processed.va.data` from UCI dataset 45. Diagnostic label values 1–4 are positive and 0 negative. Features are age, sex, chest-pain type, resting blood pressure, cholesterol, fasting glucose flag, resting ECG, maximum heart rate, exercise angina, oldpeak, ST slope, number of major vessels and thal. Question marks and the documented negative-nine missing sentinel are missing. Incomplete predictor records are retained; diagnostic labels must be valid. Categorical variables are sex, chest-pain type, fasting glucose flag, resting ECG, exercise angina, slope and thal. Imputation and category encoding are fitted on source proper training only. The four processed files retain 303, 294, 123 and 200 records. Historical diagnostic work-up and recording differences limit contemporary clinical interpretation.

For sepsis, each complete original patient file is acquired from the official PhysioNet directory or its advertised public S3 mirror and hashed. Filtering is sequential: hospital A excludes 4,499 files with fewer than 24 recorded rows and 5,663 without the ordered initial ICU clock 1–24; hospital B excludes 5,176 and 903, respectively. Remaining records require binary labels and no positive supplied label through hour 12; neither rule excludes additional otherwise eligible records in this extraction. The eligible cohorts contain 136 and 78 positive outcomes. Requiring observation through hour 24 and an observed initial clock can select patients and must not be read as prospective population validation. The supplied positive label starts six hours before the Challenge clinical onset; the endpoint is any positive supplied label at hours 13–24, not the actual onset time.

Only the first twelve hourly rows provide predictors. For each of the first 34 physiological fields, features are its observed mean, last observed value and missing fraction over those twelve rows. The static Age, Gender, Unit1, Unit2 and HospAdmTime values are the last observed within that same window. Neither SepsisLabel, ICULOS, hospital identifier nor later length of stay enters the predictor. Thus there are 107 potential numeric predictors before source-fitted preprocessing.

The external LightGBM model uses the fixed settings in Supplementary Section S9. Numeric predictors use source-fitted median imputation and standardization. Cardiac categorical predictors use source-fitted most-frequent imputation and one-hot encoding with minimum category frequency 20 and ignored unknown categories. Elastic-net SGD retains the main model settings; software versions are pinned in Supplementary Code 1, `external-validation-20261017/requirements-external.txt`. Empty source feature columns are retained by the imputer; unknown categories are ignored. No target test outcome fits preprocessing, the classifier, calibration or a model-selection rule. Source and target roles and target label prefixes are outcome-blind and patient-disjoint. External source calibration receives the ceiling of 30% of source rows, and proper training receives the remainder. The external target pool receives the ceiling of 30% of target rows, and the test receives the remainder; primary target testing instead receives the ceiling of 70%. For example, the external Cleveland target has 303 rows, partitioned into a pool of 91 and test of 212. The six institutional test counts are 212, 205, 140 and 86 for Cleveland, Hungary, VA Long Beach and Switzerland, and 9,744 and 7,121 for A-to-B and B-to-A sepsis. Table S21 reports these counts. Whole-dataset controls round the 25% test and the 30% within-development calibration upward. At 10%, cardiac label budgets are 12–30 and sepsis budgets are 1,017–1,392; class counts and overflow, not fractions alone, determine conservativeness. Source split and source Mondrian keep all source calibration labels.

The original fixed-300 stronger-adaptation comparison is not extrapolated to these external cohorts. Whole-dataset controls use an independent 25% test role and 30% calibration within development data; they are not size-matched counterfactual transport experiments.

The observation experiment uses the same uniform, outcome-only and risk-prioritized Bernoulli eligibility mechanisms as the additional outcome-blind label-observation design at the 10% planned count. An insufficient eligible set uses all its rows without redraw. Uniform sentinel labels are a counterfactual alternative collection protocol. Four methods are evaluated on each channel: ordinary and Mondrian calibration using available or sentinel labels. Repeated seeds, draws and center comparisons remain descriptive sensitivity analyses, not independent clinical replications.

Table S18: Every external institutional condition at the 10% target-label budget. Entries average draws within seed, then five seeds.

Data	Target	Model	Calibration	Coverage	Class 0	Class 1	Size
Heart	CLE	Logistic	Source Mondrian	0.7830	0.8437	0.7163	1.2594
Heart	CLE	Logistic	Source split	0.7868	0.8460	0.7226	1.2868
Heart	CLE	Logistic	Target Mondrian	0.9362	0.9157	0.9603	1.6402
Heart	CLE	Logistic	Target split	0.8979	0.9004	0.8979	1.5015
Heart	HUN	Logistic	Source Mondrian	0.9200	0.9861	0.8114	1.4117
Heart	HUN	Logistic	Source split	0.9112	0.9576	0.8341	1.3629
Heart	HUN	Logistic	Target Mondrian	0.9325	0.9366	0.9257	1.5563
Heart	HUN	Logistic	Target split	0.8872	0.9317	0.8129	1.3124
Heart	VA	Logistic	Source Mondrian	0.8200	0.6271	0.8810	1.3143
Heart	VA	Logistic	Source split	0.7900	0.6740	0.8259	1.3214
Heart	VA	Logistic	Target Mondrian	0.9177	1.0000	0.8914	1.8609
Heart	VA	Logistic	Target split	0.8669	0.7941	0.8892	1.5369
Heart	SWI	Logistic	Source Mondrian	0.8419	0.7981	0.8452	1.4070
Heart	SWI	Logistic	Source split	0.8093	0.8429	0.8083	1.4744
Heart	SWI	Logistic	Target Mondrian	0.9265	1.0000	0.9211	1.8981
Heart	SWI	Logistic	Target split	0.9330	0.8438	0.9406	1.6498
Heart	CLE	LightGBM	Source Mondrian	0.9151	0.9559	0.8683	1.4387
Heart	CLE	LightGBM	Source split	0.9057	0.9223	0.8864	1.4057
Heart	CLE	LightGBM	Target Mondrian	0.9108	0.9214	0.8987	1.4849
Heart	CLE	LightGBM	Target split	0.8696	0.8878	0.8488	1.3094
Heart	HUN	LightGBM	Source Mondrian	0.9239	0.9798	0.8318	1.3990
Heart	HUN	LightGBM	Source split	0.9259	0.9655	0.8600	1.3844
Heart	HUN	LightGBM	Target Mondrian	0.9249	0.9302	0.9146	1.5331
Heart	HUN	LightGBM	Target split	0.8960	0.9453	0.8140	1.3001
Heart	VA	LightGBM	Source Mondrian	0.7571	0.6758	0.7825	1.2714
Heart	VA	LightGBM	Source split	0.7771	0.6534	0.8169	1.2614
Heart	VA	LightGBM	Target Mondrian	0.9640	1.0000	0.9525	1.9149
Heart	VA	LightGBM	Target split	0.8911	0.7352	0.9412	1.5560
Heart	SWI	LightGBM	Source Mondrian	0.8930	0.9048	0.8925	1.5767
Heart	SWI	LightGBM	Source split	0.9116	0.8029	0.9200	1.5047
Heart	SWI	LightGBM	Target Mondrian	0.9284	1.0000	0.9232	1.9000
Heart	SWI	LightGBM	Target split	0.9288	0.8888	0.9321	1.6056
Sepsis	B to A	Logistic	Source Mondrian	1.0000	1.0000	1.0000	2.0000
Sepsis	B to A	Logistic	Source split	1.0000	1.0000	1.0000	2.0000
Sepsis	B to A	Logistic	Target Mondrian	0.9953	0.9953	1.0000	1.9952
Sepsis	B to A	Logistic	Target split	1.0000	1.0000	1.0000	2.0000
Sepsis	A to B	Logistic	Source Mondrian	0.9649	0.9647	1.0000	1.9647
Sepsis	A to B	Logistic	Source split	0.9641	0.9651	0.8040	1.7688
Sepsis	A to B	Logistic	Target Mondrian	0.9310	0.9306	1.0000	1.9304
Sepsis	A to B	Logistic	Target split	0.9377	0.9404	0.5020	1.4299
Sepsis	B to A	LightGBM	Source Mondrian	0.9111	0.9108	0.9334	1.8194

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Table S18: Every external institutional condition at the 10% target-label budget. Entries average draws within seed, then five seeds.

Data	Target	Model	Calibration	Coverage	Class 0	Class 1	Size
Sepsis	B to A	LightGBM	Source split	0.9032	0.9154	0.0000	0.9143
Sepsis	B to A	LightGBM	Target Mondrian	0.8997	0.8990	0.9512	1.8374
Sepsis	B to A	LightGBM	Target split	0.9030	0.9152	0.0000	0.9142
Sepsis	A to B	LightGBM	Source Mondrian	0.9790	0.9806	0.7081	1.6110
Sepsis	A to B	LightGBM	Source split	0.9779	0.9837	0.0000	0.9835
Sepsis	A to B	LightGBM	Target Mondrian	0.8965	0.8960	0.9899	1.8789
Sepsis	A to B	LightGBM	Target split	0.8956	0.9009	0.0000	0.9005

Table S19: All observation channels and alternative calibration protocols in the external cohorts. Conditions receive equal weight within each collection.

Data	Model	Observation	Calibration	Coverage	Class 0	Class 1	Size
Heart	Logistic	Outcome-only	Available Mondrian	0.9221	0.9498	0.9346	1.7899
Heart	Logistic	Outcome-only	Available split	0.8895	0.8782	0.8664	1.4948
Heart	Logistic	Outcome-only	Uniform Mondrian	0.9210	0.9636	0.9156	1.7383
Heart	Logistic	Outcome-only	Uniform split	0.9056	0.8705	0.8947	1.5206
Heart	Logistic	Risk-priority	Available Mondrian	0.8757	0.9826	0.8297	1.6404
Heart	Logistic	Risk-priority	Available split	0.8869	0.8699	0.8773	1.4982
Heart	Logistic	Risk-priority	Uniform Mondrian	0.9312	0.9573	0.9375	1.7674
Heart	Logistic	Risk-priority	Uniform split	0.9027	0.8688	0.8941	1.5159
Heart	Logistic	Uniform	Available Mondrian	0.9225	0.9600	0.9175	1.7337
Heart	Logistic	Uniform	Available split	0.9049	0.8748	0.8963	1.5296
Heart	Logistic	Uniform	Uniform Mondrian	0.9254	0.9595	0.9215	1.7353
Heart	Logistic	Uniform	Uniform split	0.9044	0.8777	0.8963	1.5329
Heart	LightGBM	Outcome-only	Available Mondrian	0.9268	0.9385	0.9475	1.7812
Heart	LightGBM	Outcome-only	Available split	0.8841	0.8537	0.8622	1.4157
Heart	LightGBM	Outcome-only	Uniform Mondrian	0.9296	0.9568	0.9233	1.7044
Heart	LightGBM	Outcome-only	Uniform split	0.8990	0.8657	0.8853	1.4457
Heart	LightGBM	Risk-priority	Available Mondrian	0.8635	0.9803	0.8127	1.5968
Heart	LightGBM	Risk-priority	Available split	0.8776	0.8478	0.8640	1.3843
Heart	LightGBM	Risk-priority	Uniform Mondrian	0.9413	0.9602	0.9430	1.7481
Heart	LightGBM	Risk-priority	Uniform split	0.9028	0.8677	0.8885	1.4653
Heart	LightGBM	Uniform	Available Mondrian	0.9313	0.9594	0.9229	1.7128
Heart	LightGBM	Uniform	Available split	0.9026	0.8724	0.8887	1.4624
Heart	LightGBM	Uniform	Uniform Mondrian	0.9321	0.9522	0.9291	1.7069
Heart	LightGBM	Uniform	Uniform split	0.9014	0.8742	0.8868	1.4640
Sepsis	Logistic	Outcome-only	Available Mondrian	0.9599	0.9596	1.0000	1.9595
Sepsis	Logistic	Outcome-only	Available split	0.9584	0.9598	0.7234	1.6729
Sepsis	Logistic	Outcome-only	Uniform Mondrian	0.9602	0.9599	1.0000	1.9598
Sepsis	Logistic	Outcome-only	Uniform split	0.9660	0.9673	0.7437	1.7038
Sepsis	Logistic	Risk-priority	Available Mondrian	0.9780	0.9778	1.0000	1.9778
Sepsis	Logistic	Risk-priority	Available split	0.9772	0.9785	0.7718	1.7397
Sepsis	Logistic	Risk-priority	Uniform Mondrian	0.9632	0.9630	1.0000	1.9629
Sepsis	Logistic	Risk-priority	Uniform split	0.9647	0.9661	0.7416	1.7010
Sepsis	Logistic	Uniform	Available Mondrian	0.9562	0.9558	1.0000	1.9557
Sepsis	Logistic	Uniform	Available split	0.9598	0.9614	0.7190	1.6714
Sepsis	Logistic	Uniform	Uniform Mondrian	0.9642	0.9639	1.0000	1.9639
Sepsis	Logistic	Uniform	Uniform split	0.9635	0.9649	0.7367	1.6937
Sepsis	LightGBM	Outcome-only	Available Mondrian	0.8978	0.8968	1.0000	1.8961
Sepsis	LightGBM	Outcome-only	Available split	0.8904	0.8990	0.0000	0.8983
Sepsis	LightGBM	Outcome-only	Uniform Mondrian	0.8971	0.8968	0.9476	1.8248
Sepsis	LightGBM	Outcome-only	Uniform split	0.8974	0.9061	0.0000	0.9054
Sepsis	LightGBM	Risk-priority	Available Mondrian	0.9205	0.9202	0.9567	1.8631

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Table S19: All observation channels and alternative calibration protocols in the external cohorts. Conditions receive equal weight within each collection.

Data	Model	Observation	Calibration	Coverage	Class 0	Class 1	Size
Sepsis	LightGBM	Risk-priority	Available split	0.9184	0.9273	0.0000	0.9267
Sepsis	LightGBM	Risk-priority	Uniform Mondrian	0.8974	0.8969	0.9651	1.8515
Sepsis	LightGBM	Risk-priority	Uniform split	0.8977	0.9065	0.0000	0.9057
Sepsis	LightGBM	Uniform	Available Mondrian	0.8981	0.8977	0.9541	1.8374
Sepsis	LightGBM	Uniform	Available split	0.8984	0.9071	0.0000	0.9064
Sepsis	LightGBM	Uniform	Uniform Mondrian	0.9004	0.9000	0.9542	1.8389
Sepsis	LightGBM	Uniform	Uniform split	0.9002	0.9090	0.0000	0.9083

Table S20: External channel-experiment label counts; ranges span all conditions, seeds and draws. Mondrian and ordinary split share the same samples. Short quotas are retained.

dataset	backbone	regime	method	Planned	Actual	Class 0	Class 1	Short quotas
Heart	Logistic	Outcome-only	Available Mondrian	12–30	12–30	0–29	0–14	0
Heart	Logistic	Outcome-only	Uniform Mondrian	12–30	12–30	0–24	5–19	0
Heart	Logistic	Risk-priority	Available Mondrian	12–30	12–30	0–20	9–22	0
Heart	Logistic	Risk-priority	Uniform Mondrian	12–30	12–30	0–23	6–17	0
Heart	Logistic	Uniform	Available Mondrian	12–30	12–30	0–25	5–19	0
Heart	Logistic	Uniform	Uniform Mondrian	12–30	12–30	0–23	6–20	0
Heart	LightGBM	Outcome-only	Available Mondrian	12–30	12–30	0–29	0–14	0
Heart	LightGBM	Outcome-only	Uniform Mondrian	12–30	12–30	0–24	5–19	0
Heart	LightGBM	Risk-priority	Available Mondrian	12–30	12–30	0–21	8–22	0
Heart	LightGBM	Risk-priority	Uniform Mondrian	12–30	12–30	0–23	6–17	0
Heart	LightGBM	Uniform	Available Mondrian	12–30	12–30	0–25	5–19	0
Heart	LightGBM	Uniform	Uniform Mondrian	12–30	12–30	0–23	6–20	0
Sepsis	Logistic	Outcome-only	Available Mondrian	1017–1392	1017–1392	1010–1392	0–7	0
Sepsis	Logistic	Outcome-only	Uniform Mondrian	1017–1392	1017–1392	996–1391	1–21	0
Sepsis	Logistic	Risk-priority	Available Mondrian	1017–1392	1017–1392	995–1390	2–22	0
Sepsis	Logistic	Risk-priority	Uniform Mondrian	1017–1392	1017–1392	998–1390	2–19	0
Sepsis	Logistic	Uniform	Available Mondrian	1017–1392	1017–1392	995–1391	1–22	0
Sepsis	Logistic	Uniform	Uniform Mondrian	1017–1392	1017–1392	995–1390	2–22	0
Sepsis	LightGBM	Outcome-only	Available Mondrian	1017–1392	1017–1392	1010–1392	0–7	0
Sepsis	LightGBM	Outcome-only	Uniform Mondrian	1017–1392	1017–1392	996–1391	1–21	0
Sepsis	LightGBM	Risk-priority	Available Mondrian	1017–1392	1017–1392	997–1390	2–20	0
Sepsis	LightGBM	Risk-priority	Uniform Mondrian	1017–1392	1017–1392	998–1390	2–19	0
Sepsis	LightGBM	Uniform	Available Mondrian	1017–1392	1017–1392	995–1391	1–22	0
Sepsis	LightGBM	Uniform	Uniform Mondrian	1017–1392	1017–1392	995–1390	2–22	0

Table S21: Untouched external-test predictor diagnostics, averaged over five source-model seeds. Random controls are contextual and not size-matched.

dataset	backbone	condition	n_test	Positive min	Positive max	auroc	Balanced accuracy
Heart	Logistic	CLE	212	95	103	0.7043	0.6833
Heart	Logistic	HUN	205	74	80	0.8341	0.7616
Heart	Logistic	VA	140	104	107	0.6499	0.5965
Heart	Logistic	Random	230	123	131	0.8590	0.7802
Heart	Logistic	SWI	86	79	81	0.7437	0.6608
Heart	LightGBM	CLE	212	95	103	0.8206	0.7515
Heart	LightGBM	HUN	205	74	80	0.8464	0.7533

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Table S21: Untouched external-test predictor diagnostics, averaged over five source-model seeds. Random controls are contextual and not size-matched.

dataset	backbone	condition	n_test	Positive min	Positive max	auroc	Balanced accuracy
Heart	LightGBM	VA	140	104	107	0.6413	0.6344
Heart	LightGBM	Random	230	123	131	0.8706	0.7874
Heart	LightGBM	SWI	86	79	81	0.7948	0.7447
Sepsis	Logistic	B to A	7121	88	104	0.4469	0.4914
Sepsis	Logistic	A to B	9744	45	68	0.5330	0.5353
Sepsis	Logistic	Random	6024	53	61	0.5768	0.5631
Sepsis	LightGBM	B to A	7121	88	104	0.5561	0.5000
Sepsis	LightGBM	A to B	9744	45	68	0.5965	0.5000
Sepsis	LightGBM	Random	6024	53	61	0.6005	0.5078

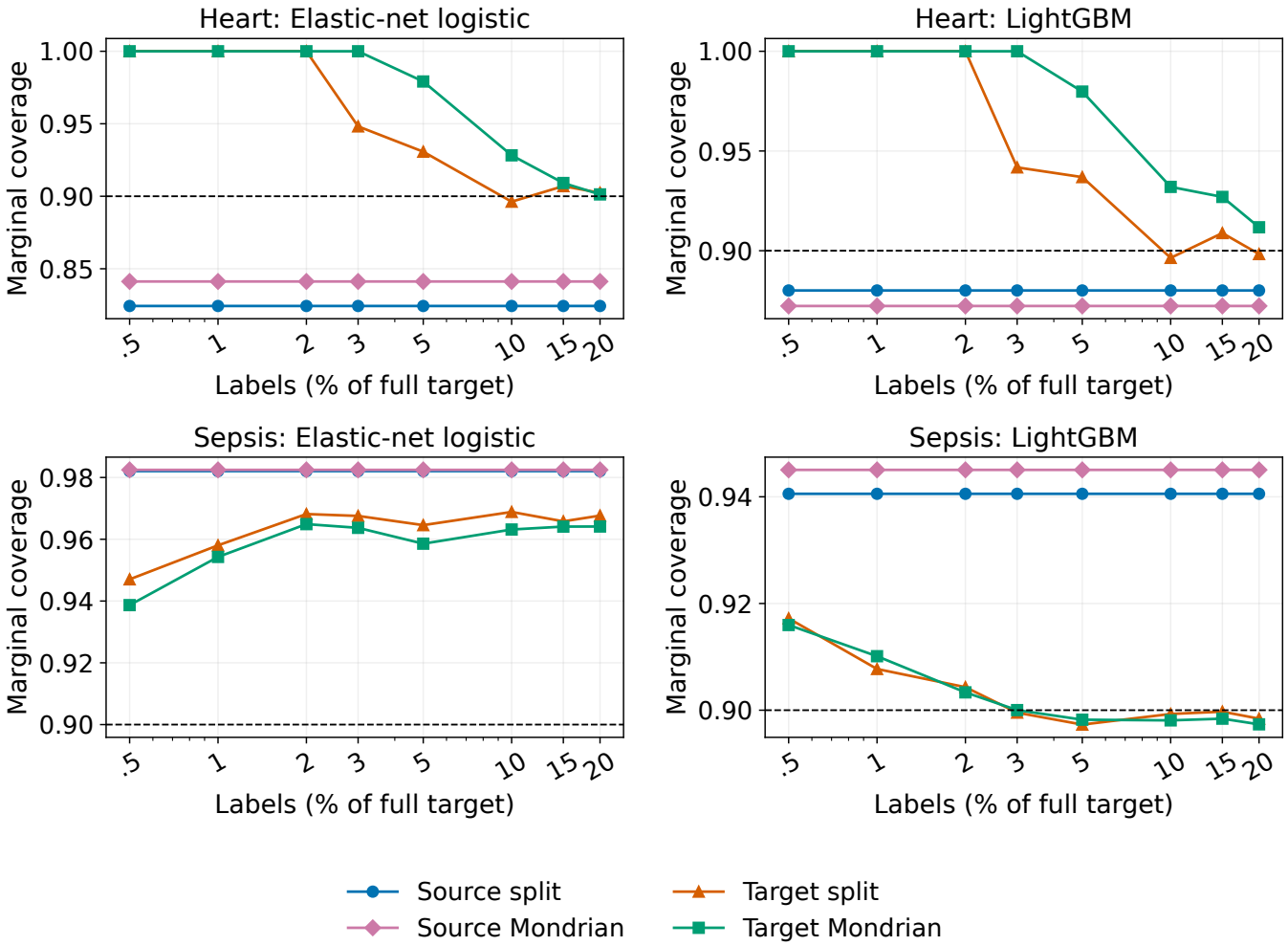


Figure S2: All eight external label fractions. Coverage first averages draws and seeds within each institutional condition, then conditions within collection. Declining conservative coverage at larger budgets need not be a deterioration. The curves retain both predictor families and every principal calibration method.

S15. Posthoc source-model diagnostic

A posthoc source-only L2 logistic sensitivity assesses model dependence associated with modest sepsis discrimination and extreme SGD probabilities. It uses source-only median imputation and standardization, LogisticRegression with C=1, lbfgs, balanced class weights and at most 5,000 iterations.

Both optimization and regularization differ from the frozen elastic-net SGD predictor, so this comparison cannot isolate an optimizer effect. Both institutional directions and whole-dataset controls use the same five seeds and unchanged patient roles; five 10% draws and all observation channels are evaluated. No hyperparameter or model is selected by target outcomes. The L2 results form a separate posthoc source-model sensitivity and are reported alongside the two prespecified predictor families.

Table S22: Posthoc source-only L2 logistic sensitivity at the unchanged 10% budget. Optimization and regularization both differ from the frozen SGD predictor. The two prespecified predictor families are reported in Table S18.

Target	Calibration	Coverage	Class 0	Class 1	Size	Source AUROC	Target AUROC
B to A	Source Mondrian	0.9625	0.9691	0.4688	1.5150	0.6198	0.4677
B to A	Source split	0.9577	0.9695	0.0805	1.0459	0.6198	0.4677
B to A	Target Mondrian	0.9040	0.9033	0.9563	1.8615	0.6198	0.4677
B to A	Target split	0.9052	0.9168	0.0444	0.9462	0.6198	0.4677
A to B	Source Mondrian	0.9439	0.9462	0.5764	1.5044	0.6259	0.5236
A to B	Source split	0.9427	0.9468	0.2610	1.1265	0.6259	0.5236
A to B	Target Mondrian	0.8964	0.8959	0.9888	1.8869	0.6259	0.5236
A to B	Target split	0.8957	0.9000	0.1894	1.0231	0.6259	0.5236

S16. Institutional result summaries

Table S23: Independent institutional cohorts and 10% target-label counts. Positive prevalence refers to the full target cohort. Class-count ranges span the 25 seed–draw combinations; both predictor families use the same uniformly sampled rows. CLE, HUN, VA and SWI denote Cleveland, Hungary, VA Long Beach and Switzerland; sepsis arrows identify source and target hospitals.

Dataset	Target	Source n	Target n	Positive (%)	Labels	Class 0	Class 1
Heart	CLE	617	303	45.87	30	14–19	11–16
Heart	HUN	626	294	36.05	29	15–24	5–14
Heart	VA	720	200	74.50	20	3–7	13–17
Heart	SWI	797	123	93.50	12	0–2	10–12
Sepsis	A to B	10174	13921	0.56	1392	1379–1390	2–13
Sepsis	B to A	13921	10174	1.34	1017	994–1010	7–23

The cardiac collection reproduces heterogeneous source-threshold transport and an acquisition–resolution tradeoff (Table S24; Figure S3). At the 10% budget, mean source split coverage across the four centers is 0.8243 with elastic-net SGD and 0.8801 with LightGBM. Ordinary target split gives 0.8963 and 0.8964, whereas target Mondrian gives 0.9282 and 0.9320, with mean set sizes 1.7389 and 1.7082. Ordinary recalibration does not reach 0.90 in every center–predictor condition. In VA Long Beach and Switzerland, the rare negative class has fewer than nine calibration labels in every 10% draw, making its Mondrian threshold infinite. The resulting conservativeness is visible alongside improved class protection, rather than counted as an unqualified success.

The sepsis collection does not reproduce source marginal undercoverage uniformly. Source split coverage is already conservative: 0.9821 for elastic-net SGD and 0.9406 for LightGBM. LightGBM target split has marginal coverage 0.8993 but positive-outcome set coverage 0.0000; target Mondrian gives marginal coverage 0.8981 and positive coverage 0.9706 at mean size 1.8582. Elastic-net target Mondrian has positive coverage 1.0000 and mean size 1.9628. Thus marginal coverage can remain

favorable while rare outcomes are unprotected, and class protection can require nearly two-label sets. Positive outcomes constitute only 1.34% and 0.56% of the eligible A and B cohorts. At 10%, positive-class rank overflow occurs in 12% of A-target and 76% of B-target calibration draws. Finite-rank conservativeness, predictor discrimination and clinical usefulness must therefore be distinguished.

Table S24: Independent institutional evaluation at the 10% budget: means over conditions within each collection after averaging five seeds and five draws. Logistic denotes the frozen elastic-net SGD predictor. Class 0 and class 1 are outcome-class set coverages, not point-classifier specificity and sensitivity. Collection means do not weight patients or independent deployments equally.

Dataset	Model	Calibration	Coverage	Class 0	Class 1	Size
Heart	Logistic	Source Mondrian	0.8412	0.8137	0.8135	1.3481
Heart	Logistic	Source split	0.8243	0.8301	0.7977	1.3614
Heart	Logistic	Target Mondrian	0.9282	0.9631	0.9246	1.7389
Heart	Logistic	Target split	0.8963	0.8675	0.8851	1.5001
Heart	LightGBM	Source Mondrian	0.8723	0.8791	0.8438	1.4215
Heart	LightGBM	Source split	0.8801	0.8360	0.8708	1.3890
Heart	LightGBM	Target Mondrian	0.9320	0.9629	0.9222	1.7082
Heart	LightGBM	Target split	0.8964	0.8643	0.8840	1.4428
Sepsis	Logistic	Source Mondrian	0.9825	0.9824	1.0000	1.9823
Sepsis	Logistic	Source split	0.9821	0.9825	0.9020	1.8844
Sepsis	Logistic	Target Mondrian	0.9632	0.9629	1.0000	1.9628
Sepsis	Logistic	Target split	0.9688	0.9702	0.7510	1.7150
Sepsis	LightGBM	Source Mondrian	0.9450	0.9457	0.8208	1.7152
Sepsis	LightGBM	Source split	0.9406	0.9495	0.0000	0.9489
Sepsis	LightGBM	Target Mondrian	0.8981	0.8975	0.9706	1.8582
Sepsis	LightGBM	Target split	0.8993	0.9080	0.0000	0.9074

The label-observation effect also varies by collection (Figure S4). For cardiac records, risk-prioritized available-label Mondrian has positive coverage 0.8297 with elastic-net SGD and 0.8127 with LightGBM; uniform sentinel Mondrian gives 0.9375 and 0.9430, with larger mean sets. The same penalty is not universal: sepsis elastic-net positive coverage is 1.0000 under both available and sentinel Mondrian; the LightGBM comparison is 0.9567 versus 0.9651. Ordinary and Mondrian alternatives and every uniform, outcome-only and risk-prioritized channel are retained in Supplementary Section S14. These results support testing label representativeness jointly with class counts and set resolution, rather than a universal advantage for sentinel sampling.

Sepsis discrimination is modest: interhospital AUROC ranges from 0.4469–0.5330 for the frozen SGD predictor and 0.5561–0.5965 for LightGBM. A posthoc source-only L2 logistic control checks sensitivity to source-model optimization and regularization without replacing these predictors or tuning on target data (Supplementary Section S15). Its ordinary target split gives marginal coverage 0.9004 and positive coverage 0.1169; target Mondrian gives positive coverage 0.9725 at mean size 1.8742. Interhospital discrimination remains weak. This exploratory diagnostic preserves the class-protection–resolution tradeoff, but does not establish an effective clinical risk predictor.

S17. Primary and additional calibration protocols

The primary protocol uses outcome-stratified role allocation and calibration sampling. Supplementary Code 2 provides its per-run results for coverage medians, condition comparisons, diagnostic

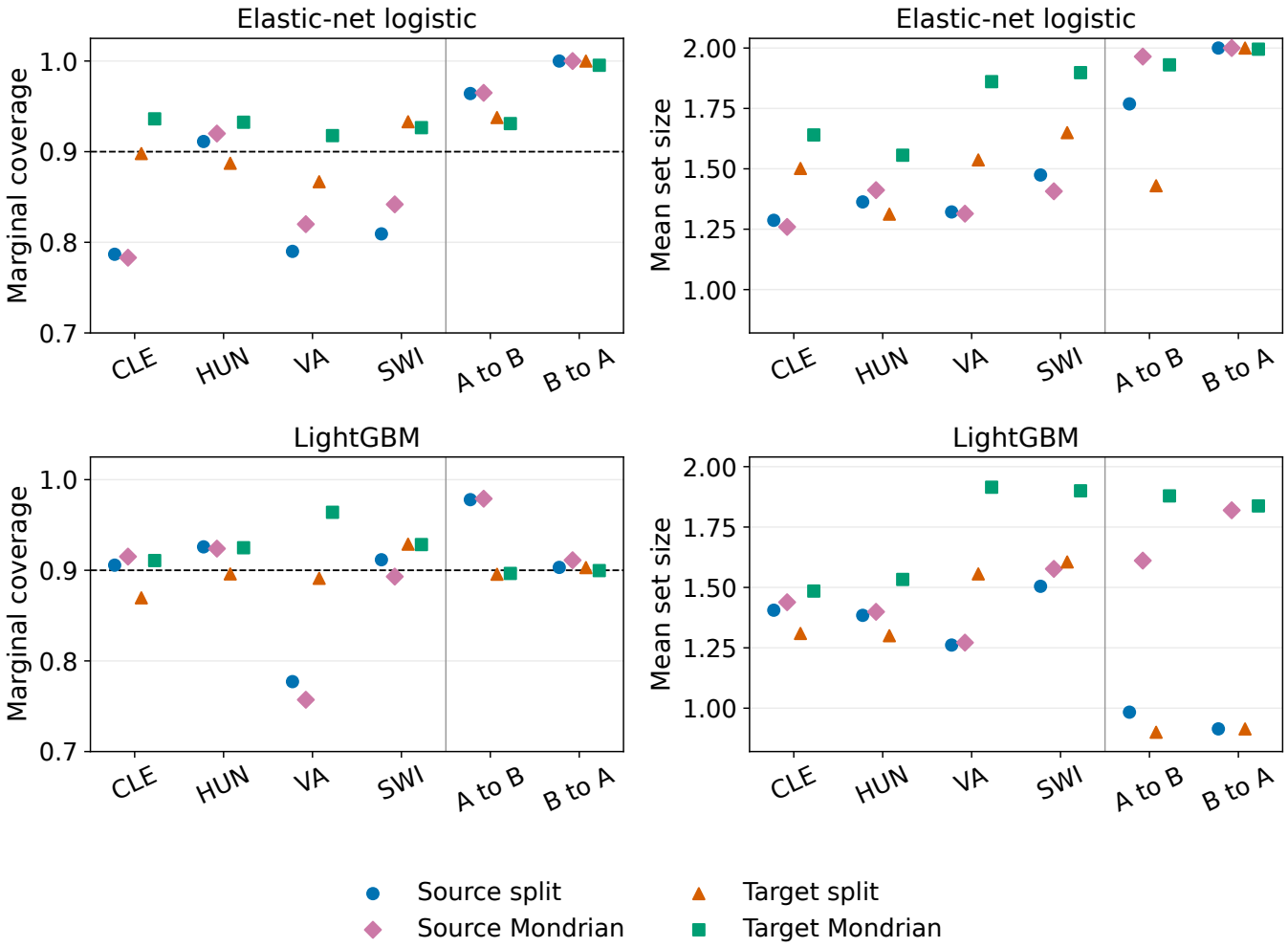


Figure S3: Every external institutional condition at 10% target labels. Points are condition means after averaging draws within each of five source-model seeds, then seeds. The left panels show marginal coverage; the right panels show the corresponding prediction-set size. The vertical divider separates four cardiac centers from the two sepsis directions. These are descriptive means without population confidence intervals. The horizontal dashed line denotes 0.90 coverage.

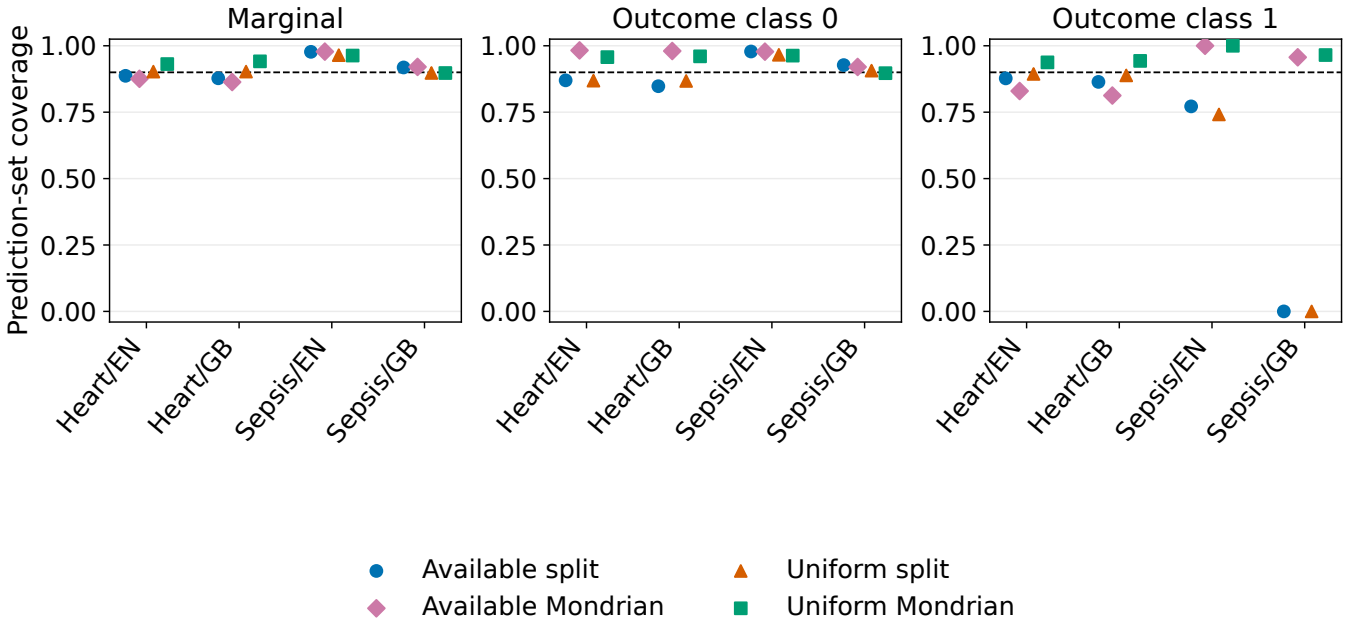


Figure S4: External risk-prioritized observation: marginal and both outcome-class coverages. EN denotes elastic-net SGD and GB denotes LightGBM. Each point equally weights the four cardiac conditions or two sepsis directions after draw and seed averaging. Uniform labels are the alternative sentinel protocol at the same planned calibration-label count, not an estimate of equal acquisition cost. The full zero-to-one scale retains the zero positive-outcome coverage of ordinary LightGBM calibration. All observation channels and actual class counts are reported in the supplement.

correlations and budget summaries. The additional outcome-blind protocol uses different role allocation, source-model refitting, a one-label rather than 100-label floor, exact finite-rank thresholds without pooled-class or empty-set fallbacks, and five calibration draws. Role allocation, label-count floor, threshold rules and draw averaging differ jointly across the two protocols. The primary ordinary thresholds use at least 100 target labels or 3,634 source labels. A class with fewer than twenty labels uses the pooled primary threshold; strict classwise guarantees are not asserted for that implementation.

Table S25: Primary and additional outcome-blind protocol summaries. These are distinct protocols and, for target methods, distinct draw-aggregation rules.

Quantity	Primary stratified	Additional outcome-blind
Random-control coverage median	0.9009	0.9000
Shifted source coverage median	0.8620	0.8616
Target split 10% coverage median	0.9011	0.9000
Target classwise 10% coverage median	0.9012	0.9003
Target classwise 10% class-gap median	0.0100	0.0127
Clipped weighting coverage median	0.8620	0.8697
Threshold Spearman correlation, sixty pairs	0.946	0.968
Threshold Spearman correlation, twelve means	0.961	0.984
All-budget target label range	100–48,000	26–48,000

Primary calibration class counts depend on the target-domain sizes, outcome-stratified allocation and model seeds. Supplementary Code 2 includes every condition, seed, fraction, label count and small-class pooling indicator. Primary ablation means of condition medians are 0.0609 and 0.0006;

means over sixty run-level shortfalls are 0.0614 and 0.0031. The two estimands summarize the same observations at different aggregation levels. The independent institutional data are excluded from both primary-dataset protocol summaries.

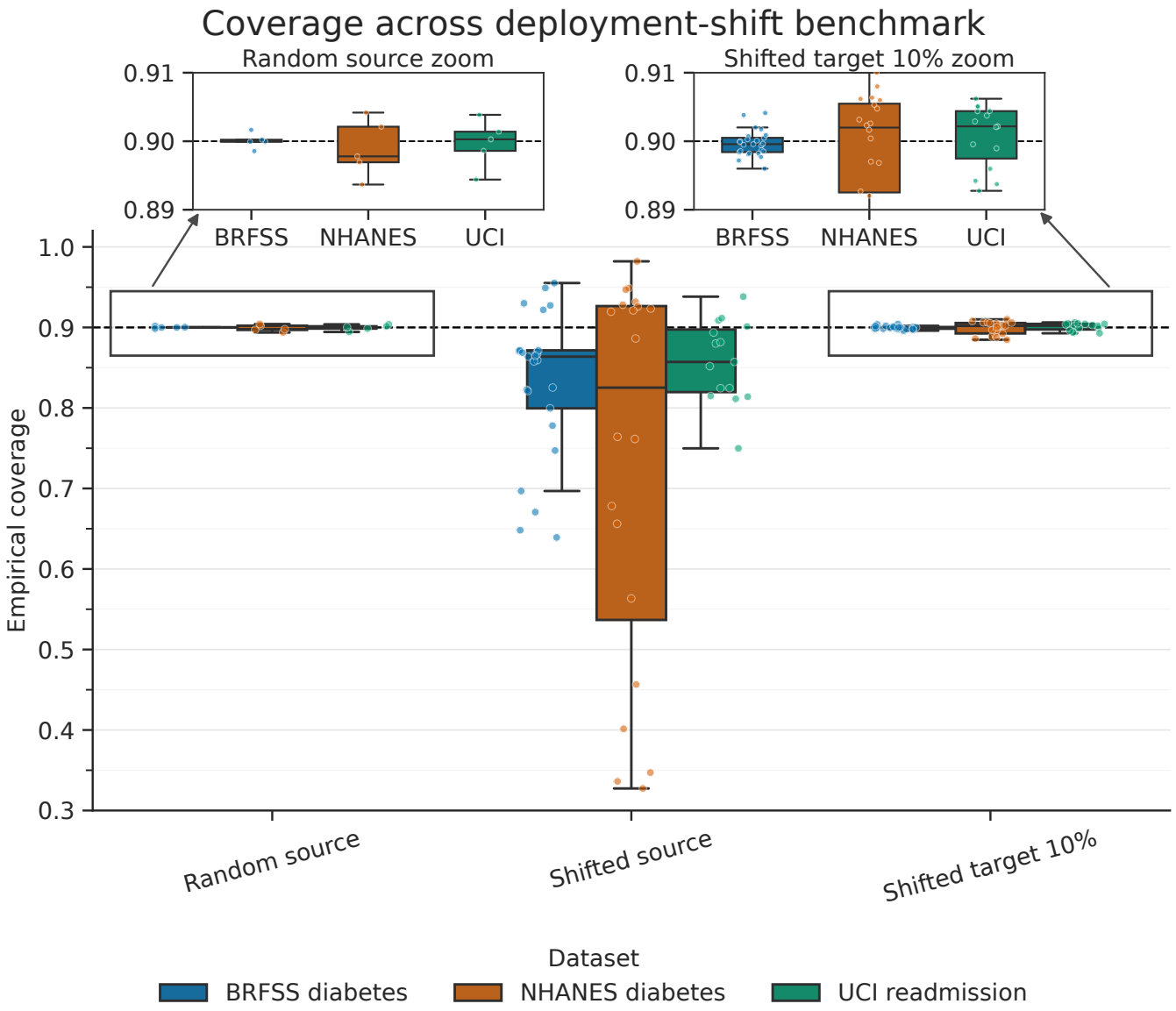


Figure S5: Outcome-blind protocol counterpart of the primary-protocol coverage comparison in Figure 2. Here random controls have median coverage 0.9000 and shifted source predictions 0.8616; five target calibration draws are averaged within condition and model seed.

S18. Detailed primary-protocol evidence

The following tables report the primary condition results, heterogeneity analyses, feature ablation, all budgets, demographic coverage, worst subgroups, descriptive tests, L2 sensitivity and score diagnostics. Nominal tests and condition-resampling intervals describe the specified related conditions rather than independent deployment populations. Pooled subgroup denominators count repeated observation appearances; their nominal Wilson ranges are not uncertainty intervals for independent individuals. Primary ablation table entries summarize medians across five seeds within condition. For demographic tables, several subgroup-CSV fields, including mean set size, are inherited from

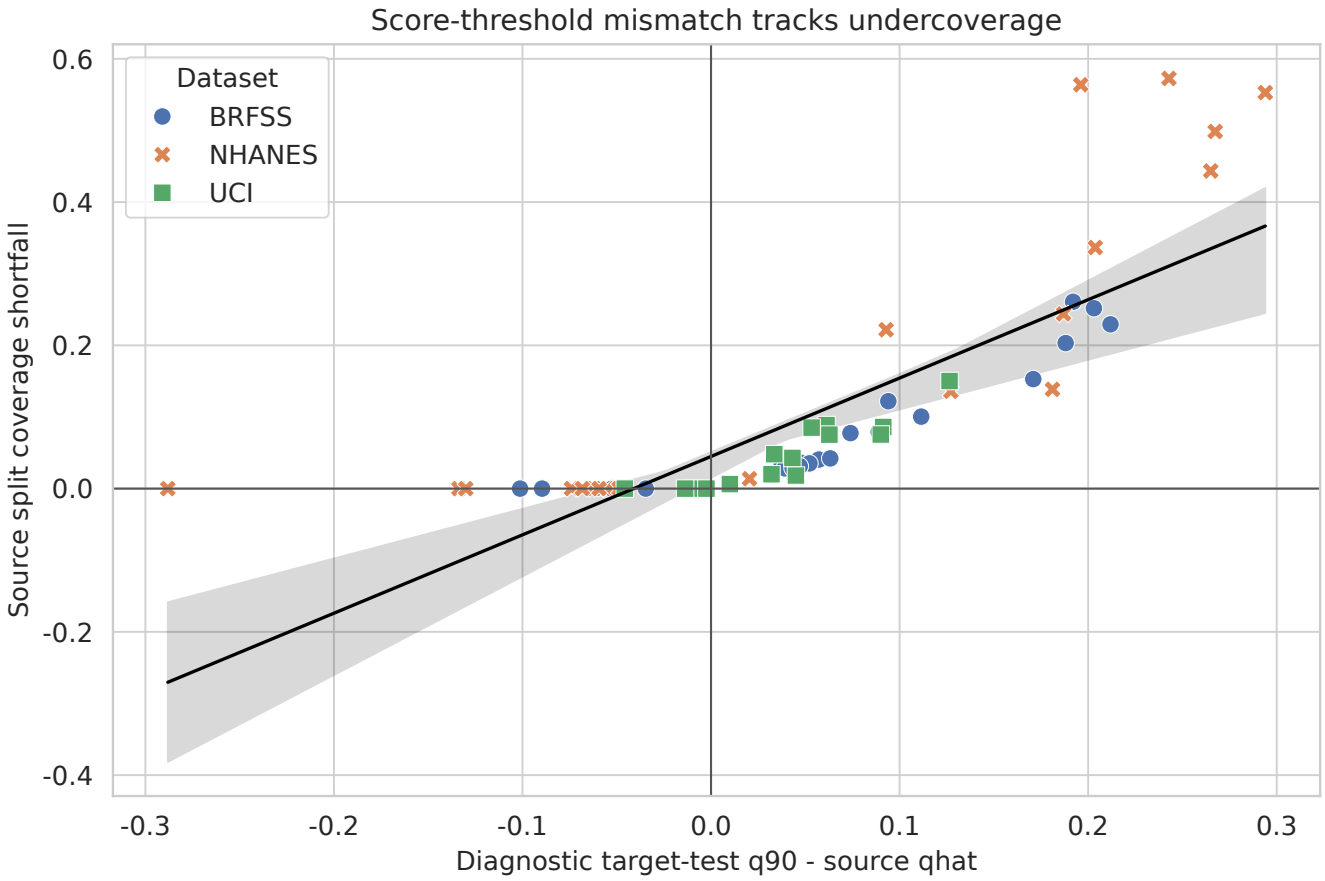


Figure S6: Outcome-blind counterpart of the primary-protocol threshold diagnostic in Figure 3. The additional protocol has descriptive correlations 0.968 across sixty pairs and 0.984 across twelve condition means.

Table S26: Additional outcome-blind protocol: condition means. OT and LR-QR use the separate fixed-300 comparison; full-source and fractional-target columns have their declared information budgets.

Condition	Source split, full	Target split 10%	Target M. 10%	OT 300	LR-QR
BRFSS Later years	0.9367	0.9001	0.9002	0.9986	0.9137
BRFSS South region	0.8681	0.8997	0.8992	0.9949	0.9761
BRFSS Age 65+	0.8093	0.9007	0.9011	0.9868	0.8918
BRFSS BMI threshold	0.6804	0.8989	0.8990	0.9358	0.9885
BRFSS Race proxy	0.8648	0.8989	0.8996	0.9949	0.9340
NHANES Later years	0.9170	0.9013	0.9013	0.9965	0.9872
NHANES Age 65+	0.3737	0.8938	0.8905	0.9333	0.7125
NHANES BMI threshold	0.6847	0.9032	0.9009	0.9618	0.8091
NHANES Race proxy	0.9459	0.8982	0.8991	0.9974	0.9477
UCI Age 70+	0.8029	0.9035	0.9029	0.9692	0.9777
UCI Race proxy	0.9106	0.8981	0.9003	0.9807	0.9034
UCI Admission source	0.8591	0.9007	0.9010	0.9722	0.9972

Table S27: Additional outcome-blind protocol: all eight fractional budgets. Coverage, per-draw absolute class gap, per-draw shortfall and set size are averaged over five calibration draws within each condition and model seed. Coverage and class-gap entries are medians over sixty such means; shortfall and size entries are means of those means.

Workflow	Fraction	Coverage	Mean shortfall	Runs < 0.90	Class gap	Size
Target Mondrian	0.005	0.9050	0.0065	19/60	0.0533	1.6184
Target Mondrian	0.010	0.9011	0.0064	26/60	0.0377	1.5225
Target Mondrian	0.020	0.9008	0.0049	24/60	0.0289	1.4813
Target Mondrian	0.030	0.9005	0.0039	26/60	0.0213	1.4751
Target Mondrian	0.050	0.8998	0.0033	31/60	0.0186	1.4657
Target Mondrian	0.100	0.9003	0.0028	29/60	0.0127	1.4606
Target Mondrian	0.150	0.9003	0.0022	29/60	0.0098	1.4607
Target Mondrian	0.200	0.8999	0.0020	30/60	0.0093	1.4604
Target split	0.005	0.9022	0.0087	26/60	0.1001	1.5465
Target split	0.010	0.9007	0.0090	26/60	0.0995	1.5340
Target split	0.020	0.9006	0.0054	27/60	0.1021	1.5372
Target split	0.030	0.9013	0.0041	25/60	0.1018	1.5368
Target split	0.050	0.9002	0.0033	28/60	0.1040	1.5361
Target split	0.100	0.9000	0.0027	29/60	0.1050	1.5341
Target split	0.150	0.9005	0.0024	27/60	0.1054	1.5341
Target split	0.200	0.9002	0.0021	30/60	0.1066	1.5340

Table S28: Additional outcome-blind protocol: absolute and class-specific label counts across twenty-five seed–draw combinations. This protocol has a one-label floor.

Condition	0.5–20% labels	10% labels	10% class 0	10% class 1
BRFSS Age 65+	727–29,069	14,534	11,375–11,501	3,033–3,159
BRFSS BMI threshold	589–23,562	11,781	8,883–9,074	2,707–2,898
BRFSS Race proxy	137–5,484	2,742	2,119–2,201	541–623
BRFSS South region	610–24,400	12,200	10,172–10,326	1,874–2,028
BRFSS Later years	1,200–48,000	24,000	20,554–20,731	3,269–3,446
NHANES Age 65+	26–1,042	521	362–384	137–159
NHANES BMI threshold	45–1,784	892	711–745	147–181
NHANES Race proxy	43–1,740	870	772–797	73–98
NHANES Later years	91–3,627	1,813	1,621–1,658	155–192
UCI Age 70+	158–6,339	3,170	2,828–2,887	283–342
UCI Admission source	191–7,658	3,829	3,430–3,533	296–399
UCI Race proxy	64–2,577	1,289	1,166–1,195	94–123

Table S29: Additional outcome-blind protocol at ten percent: equal-dataset mean coverage and set efficiency.

Method	Coverage	Size	Empty	Singleton	Two-label	Class gap
Source split	0.8066	1.3937	0.0000	0.6063	0.3937	0.2576
Source Mondrian	0.8050	1.4011	0.0000	0.5989	0.4011	0.2074
Target split	0.8998	1.5416	0.0000	0.4584	0.5416	0.1472
Target Mondrian	0.8997	1.4865	0.0000	0.5135	0.4865	0.0161

Table S30: Additional outcome-blind protocol: paired condition effects and ranges across twelve related conditions. Ranges are not confidence intervals.

Comparison	Mean	Median	Condition range
Shift minus random-control coverage error	0.1125	0.0404	[0.0106, 0.5227]
Target10 minus source shortfall	-0.1024	-0.0362	[-0.5175, 0.0059]
Clipped weighting minus source shortfall	-0.0474	-0.0000	[-0.3833, 0.0000]

Table S31: Shift-level source split and 10% target recalibration results over five seeds. Coverage target is 0.90; positive values in the last two columns indicate improvement after target-domain recalibration.

Dataset	Shift	Src cov.	Src short.	Tgt10 cov.	Tgt10 short.	Cov. gain	Short. red.
BRFSS	age_65plus	0.82746	0.07254	0.90161	0.00071	0.07415	0.07184
BRFSS	obesity	0.69018	0.20982	0.90098	0.00013	0.21079	0.20969
BRFSS	race_black_vs_white	0.86152	0.03848	0.89653	0.00462	0.03501	0.03387
BRFSS	region_south	0.86689	0.03311	0.89784	0.00228	0.03095	0.03084
BRFSS	time_future	0.94666	0.00000	0.90041	0.00087	-0.04626	-0.00087
NHANES	age_65plus	0.36446	0.53554	0.90441	0.00447	0.53995	0.53106
NHANES	obesity	0.69161	0.20839	0.90772	0.00133	0.21611	0.20706
NHANES	race_black_vs_white	0.95983	0.00000	0.89722	0.00651	-0.06260	-0.00651
NHANES	time_future	0.92694	0.00000	0.90036	0.00382	-0.02658	-0.00382
UCI	age_older70	0.78593	0.11407	0.89804	0.00325	0.11211	0.11082
UCI	emergency_admission	0.86387	0.03613	0.90146	0.00202	0.03759	0.03411
UCI	race_african_american	0.91070	0.00000	0.90143	0.00270	-0.00927	-0.00270

Table S32: Per-shift heterogeneity diagnostics over five seeds. Source shortfall and target-recalibration shortfall are reported as mean (SD); prevalence difference is target minus source positive-label prevalence.

Dataset	Shift	Source short.	Tgt10 short.	Domain AUC	Prev. diff.	Short. red.
BRFSS	age_65plus	0.07254 (0.00625)	0.00071 (0.0014186)	1.00000	0.11860	0.07184
BRFSS	obesity	0.20982 (0.02773)	0.00013 (0.00019)	1.00000	0.14200	0.20969
BRFSS	race_black_vs_white	0.03848 (0.00750)	0.00462 (0.00408)	1.00000	0.08184	0.03387
BRFSS	region_south	0.03311 (0.00248)	0.00228 (0.00213)	1.00000	0.03813	0.03084
BRFSS	time_future	0.00000 (0.00000)	0.00087 (0.0014194)	1.00000	0.00938	-0.00087
NHANES	age_65plus	0.53554 (0.02186)	0.00447 (0.00764)	0.99979	0.22631	0.53106
NHANES	obesity	0.20839 (0.05490)	0.00133 (0.00296)	0.99594	0.13397	0.20706
NHANES	race_black_vs_white	0.00000 (0.00000)	0.00651 (0.01039)	1.00000	0.01378	-0.00651
NHANES	time_future	0.00000 (0.00000)	0.00382 (0.00738)	1.00000	0.01963	-0.00382
UCI	age_older70	0.11407 (0.04195)	0.00325 (0.00403)	1.00000	0.02288	0.11082
UCI	emergency_admission	0.03613 (0.01259)	0.00202 (0.00410)	1.00000	0.00494	0.03411
UCI	race_african_american	0.00000 (0.00000)	0.00270 (0.00402)	1.00000	-0.00522	-0.00270

Table S33: Shift-defining feature ablation. The primary benchmark retained shift-defining variables when they were part of the harmonized feature space. This sensitivity drops those variables, then reruns source split conformal and 10% target-domain split recalibration on the same shifts and seeds.

Dataset	Shift	Dropped variables	Source cov.	Source shortfall	Target10 cov.	Target10 shortfall
BRFSS	time_future	`survey_year`	0.94973	0.00000	0.90083	0.00000
BRFSS	region_south	`region; _STATE`	0.86509	0.03491	0.89833	0.00167
BRFSS	age_65plus	`age80; _AGE65YR; _AGEG5YR`	0.75721	0.14279	0.90268	0.00000
BRFSS	obesity	`bmi; _BMI5CAT`	0.86926	0.03074	0.90127	0.00000
BRFSS	race_black_vs_white	`_RACE`	0.86418	0.03582	0.89810	0.00190
NHANES	time_future	`cycle_start`	0.88987	0.01013	0.90239	0.00000
NHANES	age_65plus	`RIDAGEYR`	0.60241	0.29759	0.89663	0.00337
NHANES	obesity	`BMXBMI; BMXWT; BMXHT; BMXWAIST`	0.83253	0.06747	0.91114	0.00000
NHANES	race_black_vs_white	`RIDRETH3`	0.92330	0.00000	0.90425	0.00000
UCI	age_older70	`age_mid`	0.80224	0.09776	0.90080	0.00000
UCI	race_african_american	`race`	0.91586	0.00000	0.89990	0.00010
UCI	emergency_admission	`admission_source_id; admission_mode`	0.88669	0.01331	0.90307	0.00000

the parent run; these fields are marked with a parent-run prefix and subgroup size is separately reconstructed from its singleton rate in Supplementary Code 2. Counts and prevalence are retained in the subgroup-level reference CSV.

Table S34: Target recalibration budget summaries and condition-resampling intervals. Metrics are first averaged over repeated seeds within each of the 12 non-pseudo shifts; intervals are descriptive bootstrap intervals over the related conditions. Coverage and shortfall are shown to seven decimals because target-recalibrated values cluster near the nominal 0.90 target.

Workflow	Target fraction	Coverage [95% resampling interval]	Shortfall [95% resampling interval]	Set size [95% resampling interval]
target Mondrian	0.005	0.9114872 [0.9060600, 0.9173773]	0.0042504 [0.0023801, 0.0062905]	1.5239 [1.3905, 1.6559]
target Mondrian	0.010	0.9071218 [0.8996627, 0.9165244]	0.0049986 [0.0029679, 0.0072121]	1.5076 [1.3708, 1.6378]
target Mondrian	0.020	0.9065767 [0.8989422, 0.9140200]	0.0035784 [0.0009794, 0.0070753]	1.5028 [1.3767, 1.6263]
target Mondrian	0.030	0.9064390 [0.9019230, 0.9115724]	0.0024920 [0.0008861, 0.0049595]	1.5178 [1.4304, 1.6107]
target Mondrian	0.050	0.9024351 [0.9002004, 0.9048128]	0.0027715 [0.0011186, 0.0047108]	1.4737 [1.3670, 1.5865]
target Mondrian	0.100	0.9016239 [0.8998447, 0.9038396]	0.0022966 [0.0014673, 0.0033653]	1.4724 [1.3686, 1.5767]
target Mondrian	0.150	0.9001028 [0.8991251, 0.9011997]	0.0018705 [0.0014270, 0.0023255]	1.4625 [1.3607, 1.5656]
target Mondrian	0.200	0.9006558 [0.8997361, 0.9017660]	0.0019271 [0.0010756, 0.0030141]	1.4648 [1.3614, 1.5702]
target split	0.005	0.9117343 [0.9078645, 0.9153644]	0.0032114 [0.0020457, 0.0044247]	1.5799 [1.4342, 1.7111]
target split	0.010	0.9034130 [0.8982028, 0.9090780]	0.0057935 [0.0032195, 0.0085896]	1.5587 [1.4150, 1.6892]
target split	0.020	0.9026146 [0.8969622, 0.9083803]	0.0044372 [0.0021854, 0.0073017]	1.5556 [1.4063, 1.6863]
target split	0.030	0.9033543 [0.8990994, 0.9078728]	0.0033921 [0.0015417, 0.0058678]	1.5581 [1.4166, 1.6888]
target split	0.050	0.9008570 [0.8993279, 0.9024780]	0.0031758 [0.0014918, 0.0050794]	1.5496 [1.4038, 1.6791]
target split	0.100	0.9006679 [0.8990593, 0.9024783]	0.0027240 [0.0017303, 0.0037797]	1.5500 [1.4029, 1.6821]
target split	0.150	0.8994662 [0.8985909, 0.9005133]	0.0023149 [0.0016884, 0.0030416]	1.5470 [1.4052, 1.6784]
target split	0.200	0.9000574 [0.8989695, 0.9012628]	0.0024310 [0.0014630, 0.0034374]	1.5481 [1.4065, 1.6767]

Table S35: Descriptive demographic and operational subgroup audit beyond class labels. Rows report the worst pooled subgroup value for each axis and workflow across the 12 non-pseudo shifts, its denominator, a nominal Wilson range computed for that pooled value, and the coverage range across values. Counts are pooled observation appearances across related conditions and repeated seeds, not unique individuals. Nominal Wilson ranges ignore this dependence and serve as arithmetic descriptions, not population confidence intervals or subgroup-validity guarantees.

Subgroup axis	Workflow	Worst value	Worst cov. [nominal range]	Worst n	Range	Values
admission source	source	non_emergency	0.8357 [0.8329, 0.8385]	67627	0.0073	2
admission source	target split 10%	emergency_room	0.8975 [0.8963, 0.8988]	222428	0.0112	2
admission source	target Mondrian 10%	emergency_room	0.8970 [0.8958, 0.8983]	222428	0.0106	2
age	source	65_plus	0.8016 [0.8009, 0.8023]	1160093	0.1271	5
age	target split 10%	65_plus	0.8630 [0.8624, 0.8637]	1160093	0.0747	5
age	target Mondrian 10%	65_plus	0.8668 [0.8662, 0.8674]	1160093	0.0676	5
obesity	source	obese	0.7542 [0.7534, 0.7551]	1013910	0.1691	2
obesity	target split 10%	obese	0.8488 [0.8481, 0.8494]	1013910	0.0884	2
obesity	target Mondrian 10%	obese	0.8511 [0.8504, 0.8518]	1013910	0.0846	2
race proxy	source	unknown	0.8166 [0.8150, 0.8182]	222291	0.0887	10
race proxy	target split 10%	black	0.8658 [0.8646, 0.8669]	340110	0.0819	10
race proxy	target Mondrian 10%	black	0.8702 [0.8690, 0.8713]	340110	0.0700	10
region	source	territory_or_unknown	0.8118 [0.8073, 0.8163]	28659	0.0643	5
region	target split 10%	territory_or_unknown	0.8573 [0.8532, 0.8613]	28659	0.0632	5
region	target Mondrian 10%	territory_or_unknown	0.8583 [0.8542, 0.8623]	28659	0.0605	5
sex	source	female	0.8463 [0.8457, 0.8470]	1126031	0.0138	3
sex	target split 10%	unknown	0.8844 [0.8835, 0.8854]	428586	0.0233	3
sex	target Mondrian 10%	unknown	0.8858 [0.8849, 0.8868]	428586	0.0207	3

Table S36: Worst descriptive subgroup coverage by deployment shift. For each shift and workflow, subgroup rows are pooled over the five repeated seeds and the lowest subgroup value is reported as axis=value; coverage; denominator. These summaries localize residual subgroup undercoverage but do not imply subgroup-conditional validity.

Dataset	Shift	Source	Target split 10%	Target Mondrian 10%
BRFSS	age_65plus	obesity=obese; 0.6911; n=130425	obesity=obese; 0.7937; n=130425	obesity=obese; 0.8141; n=130425
BRFSS	obesity	age=65_plus; 0.5439; n=130367	age=65_plus; 0.8299; n=130367	age=65_plus; 0.8551; n=130367
BRFSS	race_black_vs_white	age=65_plus; 0.7446; n=27286	age=65_plus; 0.8028; n=27286	age=65_plus; 0.8235; n=27286
BRFSS	region_south	obesity=obese; 0.7681; n=128179	obesity=obese; 0.8106; n=128179	obesity=obese; 0.8186; n=128179
BRFSS	time_future	obesity=obese; 0.8896; n=244128	obesity=obese; 0.8120; n=244128	obesity=obese; 0.8050; n=244128
NHANES	age_65plus	sex=female; 0.3304; n=9321	obesity=obese; 0.8337; n=6229	obesity=obese; 0.8684; n=6229
NHANES	obesity	age=65_plus; 0.4498; n=6120	age=65_plus; 0.7294; n=6120	age=65_plus; 0.7598; n=6120
NHANES	race_black_vs_white	age=65_plus; 0.8916; n=3663	age=65_plus; 0.6757; n=3663	age=65_plus; 0.6787; n=3663
NHANES	time_future	age=65_plus; 0.7204; n=9680	age=65_plus; 0.6351; n=9680	age=65_plus; 0.6258; n=9680
UCI	age_older70	admission=emergency_room; 0.7707; n=61421	admission=emergency_room; 0.8906; n=61421	admission=emergency_room; 0.8923; n=61421
UCI	emergency_admission	age=older70; 0.8144; n=61147	age=older70; 0.8683; n=61147	age=older70; 0.8657; n=61147
UCI	race_african_american	age=older70; 0.8759; n=13702	age=older70; 0.8609; n=13702	age=older70; 0.8598; n=13702

Table S37: Condition-level descriptive paired Wilcoxon comparisons. Each of the 12 related conditions contributes one mean shortfall difference after seed averaging; nominal p values do not establish inference over independent deployments.

Backbone	Source short.	Tgt10 short.	Median diff.	p
Elastic-net SGD	0.1040	0.0027	-0.03399	0.020996
L2 SGD	0.1079	0.0025	-0.03888	0.020996

Table S38: Predictive-backbone sensitivity over the same 60 non-pseudo shift-run pairs. Values are medians over shift-run pairs. The L2-SGD backbone changes classifier regularization while preserving preprocessing, splitting, and conformal protocol.

Backbone	Workflow	Coverage	Shortfall	Set size	Class gap
Elastic-net SGD	source split	0.8620	0.0380	1.3811	0.1430
Elastic-net SGD	target split 10%	0.9011	0.0000	1.6147	0.1044
L2 SGD	source split	0.8597	0.0403	1.3392	0.1634
L2 SGD	target split 10%	0.9001	0.0000	1.5025	0.1162

Table S39: Direct score-threshold mechanism diagnostics. Source qhat is the source-calibration split conformal threshold; target oracle q90 is the diagnostic 90th percentile of target-test nonconformity scores computed using held-out labels.

Dataset	Shift	Source qhat	Target q90	Q gap	Source short.	Tgt10 qhat
BRFSS	age_65plus	0.7147	0.8001	0.0854	0.0725	0.8021
BRFSS	obesity	0.7276	0.9091	0.1815	0.2098	0.9098
BRFSS	race_black_vs_white	0.7379	0.7876	0.0496	0.0385	0.7825
BRFSS	region_south	0.7171	0.7655	0.0484	0.0331	0.7624
BRFSS	time_future	0.7411	0.6549	-0.0862	0.0000	0.6555
NHANES	age_65plus	0.7809	0.9812	0.2002	0.5355	0.9813
NHANES	obesity	0.7104	0.8924	0.1820	0.2084	0.9005
NHANES	race_black_vs_white	0.7817	0.5768	-0.2049	0.0000	0.5693
NHANES	time_future	0.7473	0.6749	-0.0724	0.0000	0.6777
UCI	age_older70	0.7330	0.8100	0.0769	0.1141	0.8086
UCI	emergency_admission	0.6972	0.7326	0.0354	0.0361	0.7337
UCI	race_african_american	0.6717	0.6588	-0.0128	0.0000	0.6589

Table S40: Exploratory shift-level mechanism diagnostics. Correlations are descriptive over the 12 deployment shifts and are not used as confirmatory tests.

Diagnostic	Outcome	Spearman r	p
Domain AUC	Source shortfall	-0.5800	0.04804
Domain AUC	Shortfall reduction	-0.5698	0.05309
|prevalence diff.|	Source shortfall	0.8258	0.00094
|prevalence diff.|	Shortfall reduction	0.7413	0.00580

S19. Additional comparator implementations and data preparation

S19.1. Additional outcome-blind protocol

A separate protocol tests sampling and implementation sensitivity. Role permutations use neither outcomes nor features. Source training/calibration remains 70%/30% and target pool/test remains 30%/70%; the model is refitted on these roles. Target budgets use $\max(1, \mathrm{round}(fN_T))$ labels and five nested calibration draws per model seed. The exact finite-rank threshold is

$$q_n^{(B)} = \begin{cases} s_{(k_n)}, & 1 \le k_n \le n, \\ +\infty, & n = 0 \text{ or } k_n > n. \end{cases} \tag{S1}$$

There is no argmax completion of empty sets; strict Mondrian calibration applies this rule within each class. Validity requires the relevant calibration/test exchangeability. This outcome-blind protocol supplies the additional fixed-300 stronger-method comparisons, label-observation experiment, LightGBM and adult-NHANES checks. Its fractional-budget sensitivity is reported separately in Supplementary Sections S1–S12 and S17. The protocol comparison jointly assesses sampling and implementation sensitivity. Independent institutional experiments have their own cohort roles, described in Supplementary Sections S14–S16.

S19.2. Stronger comparators and reproduction details

Within the additional outcome-blind protocol, a separate comparison fixes five draws of 300 source calibration rows and 300 target-pool rows for each of the 60 condition–seed pairs. Source split on 300 rows controls source sample size; full source split and Mondrian remain contextual references. Target split and target Mondrian reveal the outcomes of the 300 target rows. Unlabeled-target methods receive their features or probabilities without outcomes. LR-QR additionally receives 300 disjoint unlabeled source calibration rows. Thus the comparison fixes several shared sample sizes, but does not equate label availability or LR-QR's total inputs. It remains separate from the fractional-budget experiment.

The additional fixed-300 weighted-score sets use the first sorted score whose cumulative calibration mass reaches $0.9(W + w_*)$, where W is total calibration weight and w_* is candidate test-point mass at infinity. If the required mass exceeds W, the threshold is infinite; membership includes ties. For label-shift weighting, source predicted classes use $p_1 \ge 0.5$. We solve the 2×2 source confusion system against target predicted-class frequencies by minimum-norm least squares, including singular

systems. Estimated target priors are clipped below at 10^{-6} and renormalized; ratios to empirical source priors are clipped to $[0.05, 20]$, with an infinite ratio for an absent source class clipped to 20. Calibration weights are class-specific ratios, and w_* is the candidate class's ratio, giving separate thresholds for labels 0 and 1. This bounded plug-in construction can be inaccurate when the confusion system is ill-conditioned or label shift does not hold.

Score-space weighting fits a logistic domain classifier ($C = 1$, at most 2,000 iterations) on a degree-three expansion of logit probability, with probabilities clipped to $[10^{-6}, 1 - 10^{-6}]$ and features standardized using both unlabeled source and target samples. Its odds, adjusted for the source/target sample counts, are clipped to $[0.05, 20]$. Each source calibration score receives its probability-specific ratio, and w_* is the ratio at the test probability. The resulting point-specific threshold is shared by the two candidate labels. This is a score-space estimator, not full-feature density-ratio adaptation.

Our independent optimal-transport (OT) implementation uses the free-form weighted-CDF objective in Correia and Louizos (Correia and Louizos, 2025). The auxiliary target distributions are the empirical CDFs of $\min(p_1, 1 - p_1)$ and $\max(p_1, 1 - p_1)$, denoted $F_{\min}$ and $F_{\max}$. For 300 true source calibration scores, weights v_i sum to one and define their empirical CDF F_v and Gaussian-kernel density f_v. We minimize

$$\frac{1}{2}\int_0^1 f_v(t)\Big[|F_v(t) - F_{\max}(t)| + |F_v(t) - F_{\min}(t)| + F_{\min}(t) - F_{\max}(t)\Big]\, dt. \tag{S2}$$

The kernel bandwidth is $\max(0.01, \mathrm{SD}(s_i)300^{-1/5})$, using sample standard deviation. Gaussian tails are not renormalized at the interval boundaries. The integral uses trapezoidal quadrature on 401 uniform points; all CDFs are unsmoothed. Softmax logits start at zero and use 4,000 float32 CPU Adam steps at learning rate 0.001. For prediction-set thresholds, weights become $300v_i$ (mean one), and $w_* = 1$. Thus the test mass has a specified scale; this construction does not estimate a full-feature target density ratio. The published implementation uses reflected boundary KDE, classification bandwidth 0.1 times Scott's rule, and free-form weights initialized uniformly on $[0, 1]$ then mapped to log space. Its free-form threshold normalizes the original unnormalized calibration weights together with a unit test weight (Appendix C.1.1). Here kernels are unreflected, bandwidth is unscaled Scott's rule with a floor, initialization is uniform in normalized weights, and prediction weights have a fixed mean of one before adding unit test mass. The last choice fixes the relative infinity mass and is not asserted to be equivalent to the published normalization. The controls concern the displayed discretized construction. These estimates concern this specified variant, rather than an exact reproduction of the published experimental settings.

For LR-QR (Joshi et al., 2025), preprocessing is fitted on the 300 labeled and 300 additional unlabeled source rows. Whitened PCA fitted on the latter uses at most 50 components, padded with zeros to 50, plus an intercept; unknown categories use the source-supported encoding. The empirical objective is

$$\widehat{E}_1[\ell_\alpha(h(X), s(X, Y))] + \lambda\widehat{E}_3[\beta^2 h(X)^2] - 2\lambda\widehat{E}_2[\beta h(X)] + 10^{-8}\|\theta_{\text{nonintercept}}\|_2^2, \tag{S3}$$

where $h(x) = x^\top\theta$ and samples 1, 2, and 3 are labeled source, unlabeled target, and additional unlabeled source. The pinball loss is $\ell_\alpha(h, s) = (1 - \alpha)(s - h)_+ + \alpha(h - s)_+$. Ten evenly spaced regularization values from $\lambda^*/10$ to λ^*, where $\lambda^* = (n_1^{-1}/(n_2^{-1} + n_3^{-1}))^{1/3}$, are compared by three-

fold held-out gradient norm. CPU Adam uses learning rate 0.02, 1,800 steps per cross-validation fit and 8,000 for final fitting, $\beta \in [0.05, 20]$, and global gradient clipping at norm 100 in fixed batches of 20 problems. Final starts $\beta = 0.25, 1, 4$ compete by the complete empirical objective in Equation S3, including the unlabeled-source, unlabeled-target and ridge terms. Prediction-set membership is $s(x, y) \leq h(x)$. Auxiliary covariates are distinct from the predictor inputs. Neither optimizer selection nor tuning uses target test outcomes.

We compare lower and increased optimization budgets on all five draws for seed 0 in every condition, plus seeds 1–4 for NHANES age 65+: 80 problems per method. OT includes two logit perturbations with standard deviation 0.2 and an additional 8,000-step check; LR-QR includes starts $\beta = 0.1, 2, 10$. All tested settings are reported with objective and gradient diagnostics. Main results use the increased budgets for all 300 draws; sensitivity results are separate from those full averages. Independently solved two-support OT and intercept-only LR-QR controls test objective fidelity (Supplementary Section S7). These controls and finite optimizations do not certify global optima for the health-data fits.

Under the additional outcome-blind protocol, the source-domain controls have calibration/test exchangeability by random partition; the target-calibrated methods use random subsets of the same target domain, with analogous within-class exchangeability for Mondrian. These statements apply to the selected computational cohorts, not future clinical observations. None of the twelve source–target conditions is established to be pure covariate shift or pure label shift. In particular, deterministic age or BMI partitions can exclude target covariate values from source support. Temporal and geographic conditions also leave $P(Y \mid X)$ and $P(X \mid Y)$ invariance unverified. Source-based adaptation results therefore measure behavior on these possibly mixed shifts; they do not test improvements over theory under verified matching assumptions.

BRFSS annual caps use deterministic sampling within outcome strata with outcome-proportional quotas, before either protocol's separate allocation of training, calibration and test roles. BRFSS uses diabetes item DIABETE4 when populated, otherwise DIABETE3; responses 1 and 3 define positive and negative outcomes. Health-day responses 88 become zero; invalid day values are missing. BMI is rescaled from _BMI5 and values outside 10–80 are missing. The frozen race-field priority is _RACE, _PRACE1, _MRACE1, _RACE_G1, _RACEGR3, _RACEG21, then SEXVAR; the last fallback is not used by the input releases. The rule is reproduced explicitly rather than claimed as validated questionnaire harmonization. NHANES uses left joins starting from DEMO; unanswered diabetes outcomes are excluded, while unavailable predictors are imputed using source training data. UCI diagnosis codes are mapped to broad groups; patient and encounter identifiers and the outcome are excluded from model inputs.

The additional outcome-blind role generator uses SHA-256-derived permutation seeds from dataset, condition, model seed, and domain. Budget draws use separate permutations, nested prefixes, and no class balancing. Dependencies, CPU reproduction commands, and numerical-verification details are supplied in the support. Provider observations and model predictions are regenerated locally.

S20. Additional protocol and empirical analyses

S20.1. Protocol sensitivity summaries

The separately defined outcome-blind protocol gives median source coverage 0.9000 in random controls and 0.8616 under shift, with 10% ordinary and strict Mondrian target medians 0.9000 and 0.9003. Its corresponding primary-condition threshold correlations are 0.968 across sixty pairs and 0.984 across twelve means. The primary-protocol results above and these additional results are reported side by side in Supplement S17. Both protocols cover the same twelve conditions and eight budgets; changes in allocation, refitting, label-count floor, threshold rules and draw aggregation prevent attributing their difference to a single factor.

S20.2. Predictor-family and adult-cohort sensitivity summaries

The primary predictor is an elastic-net logistic SGD model; score-distribution differences may depend on its representation and fitted probabilities. Within the additional outcome-blind protocol, a different-family LightGBM sensitivity on its fixed roles and 10% target draws yielded equal-dataset source coverage 0.8314 and target Mondrian coverage 0.8999, mean size 1.4789 and class gap 0.0146. Adult NHANES sensitivity, excluding children and requiring observed BMI for its BMI condition, used five draws of 300 target labels per condition–seed pair and yielded target Mondrian coverage 0.9010; this fixed-label sensitivity is separate from the primary-protocol 10% comparison. These checks extend the qualitative comparison to a nonlinear predictor and an adult-only cohort (Supplementary Tables S5 and S10). The stronger adaptation comparison remains specific to the elastic-net predictor.

S20.3. Calibration-label observation mechanisms

Clinical outcome observation may depend on follow-up and on which records receive review. The benchmark therefore also evaluates a maintenance decision: how outcome-class calibration using routinely available target labels compares with calibration using a representative target sample. A controlled label-observation experiment compares these choices at fixed calibration-label counts, separately from the primary fractional-budget analysis.

Target-label observation experiment. We use the sixty frozen condition–seed predictors, target pools and test roles from the additional outcome-blind protocol. For each pool, planned calibration counts are 100 or 300, with five draws. Three observation mechanisms have relative weights $w_i = 1$ (uniform), $w_i = 0.2$ for positive outcomes and 1 otherwise (outcome-dependent availability), or $w_i = 0.1 + 0.9p_1(x_i)$ (risk-prioritized review). The outcome-dependent mechanism simulates missing outcomes retrospectively, using the recorded labels to construct the observation process. For each mechanism and count n, independent Bernoulli eligibility has probability $a_i = \min(1, cw_i)$, with c chosen so $\sum_i a_i = 2n$. We uniformly select n eligible rows; an insufficient quota would use all eligible rows and be reported without redrawing. All quotas are feasible in the observed runs. An alternative sentinel protocol draws n rows uniformly from the full target pool, independently of outcomes and

scores. Each alternative supplies n calibration labels; total acquisition costs depend on the collection process.

Ordinary and Mondrian target calibration are evaluated on each label source. A prespecified protocol rule uses available-label Mondrian calibration for uniform or outcome-only observation, and sentinel Mondrian calibration for risk-prioritized observation. The rule selects the calibration procedure from the known sampling design, without using test outcomes. Outcome-only selection preserves feature sampling within each outcome class, whereas score-based selection can change that conditional distribution. The primary endpoints are both outcome-class coverages and positive-class shortfall; set size and singleton errors quantify efficiency. These controlled scenarios isolate the effect of label observation on calibration.

Results. The controlled observation experiment distinguishes two reasons why a local label set can be unrepresentative (Table S41). With 300 risk-prioritized labels, available-label Mondrian calibration has equal-dataset marginal coverage 0.9318 but positive-outcome coverage 0.8527. Positive coverage is below 0.90 in all twelve condition means (range 0.8149–0.8952). The same label count drawn through the uniform sentinel protocol gives positive coverage 0.9092 and marginal coverage 0.9023, with condition positive coverage ranging from 0.8959 to 0.9254; one condition mean remains below 0.90. The increase in marginal coverage under selective observation can therefore conceal reduced protection of positive outcomes.

Under outcome-dependent availability, Mondrian positive coverage is 0.9611 at 300 labels, but mean set size is 1.6966, compared with 1.5125 for sentinel Mondrian. Small class counts explain part of this conservativeness. With risk-prioritized labels, ordinary split calibration also has positive coverage 0.9222, but larger sets (1.6454) than sentinel Mondrian (1.5124). The protocol rule links calibration-label source and class conditioning to the known observation mechanism. Supplementary Section S13 reports every method, label/class count and condition; Figure S1 displays the complete risk-prioritized comparison. These comparisons reuse the benchmark's fixed test roles.

The label-observation comparison gives a specific methodological consequence: outcome-class thresholding can address outcome-only sampling differences, but does not by itself address risk-prioritized sampling within an outcome class. A representative sentinel sample changes that calibration-data assumption. The appropriate protocol therefore depends on how labels enter the system, not only on an aggregate coverage estimate.

S20.4. Institutional overview and scope

Institutional results. Additional institutional evaluations on 920 cardiac diagnostic records and 24,095 eligible sepsis patients extend the assessment beyond the three primary datasets (Janosi et al., 1989; Reyna et al., 2019, 2020; Pollard et al., 2026). The cardiac comparisons support heterogeneous coverage repair and label-observation effects. In sepsis, source coverage can already be conservative, and near-nominal marginal coverage can coexist with zero positive-outcome set coverage; outcome-class calibration improves rare-class protection at the cost of large sets. Interhospital discrimination remains weak and sentinel benefits vary by collection and predictor. Complete cohort definitions, methods and favorable and unfavorable results are reported in Supplementary Sections S14–S16, Tables S18–S24 and Figures S2–S4.

Table S41: Controlled target-label observation experiment: equal-dataset means. C_1 and C_0 denote positive- and negative-outcome set coverage, not point-classifier sensitivity or clinical benefit. Available split and available Mondrian use available labels; sentinel Mondrian uses uniform labels. Each method row aligns both class coverages with its own mean set size. The last column identifies the row selected by the known-design protocol rule: available Mondrian for uniform/outcome-only observation and sentinel Mondrian for risk-prioritized observation. Label counts concern calibration inputs, not total acquisition cost.

Labels	Observation channel	Method	C_0	C_1	Size	Rule selects
100	Outcome-dependent	Available split	0.9008	0.8671	1.5475	
100	Outcome-dependent	Available Mondrian	0.9047	0.9952	1.8567	Yes
100	Outcome-dependent	Sentinel Mondrian	0.9048	0.9363	1.6181	
100	Risk-prioritized	Available split	0.9308	0.9196	1.6418	
100	Risk-prioritized	Available Mondrian	0.9456	0.8736	1.5285	
100	Risk-prioritized	Sentinel Mondrian	0.9028	0.9348	1.6105	Yes
100	Uniform	Available split	0.8954	0.8798	1.5459	
100	Uniform	Available Mondrian	0.9031	0.9376	1.6221	Yes
100	Uniform	Sentinel Mondrian	0.9060	0.9332	1.6049	
300	Outcome-dependent	Available split	0.9000	0.8674	1.5419	
300	Outcome-dependent	Available Mondrian	0.9015	0.9611	1.6966	Yes
300	Outcome-dependent	Sentinel Mondrian	0.9011	0.9109	1.5125	
300	Risk-prioritized	Available split	0.9322	0.9222	1.6454	
300	Risk-prioritized	Available Mondrian	0.9449	0.8527	1.4871	
300	Risk-prioritized	Sentinel Mondrian	0.9013	0.9092	1.5124	Yes
300	Uniform	Available split	0.8951	0.8822	1.5441	
300	Uniform	Available Mondrian	0.9014	0.9122	1.5161	Yes
300	Uniform	Sentinel Mondrian	0.9023	0.9043	1.5037	

The external cardiac centers are small historical cohorts with differing diagnostic work-up; the sepsis endpoint uses supplied Challenge labels and conditions on recorded observation through hour 24 and the initial ICU clock. Weak interhospital discrimination and rare outcomes constrain the informativeness of its prediction sets.

References

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